

Implicit Incentive Provision with Misspecified Learning

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Abstract

We study implicit incentive provision with misspecified Bayesian learning, where an agent is assessed by an evaluator and rewarded by the market. The agent’s outcome depends on their innate ability, costly effort—whose effectiveness is governed by a productivity parameter—and noise. The market infers the agent’s ability from observed outcomes and rewards them accordingly. The evaluator conducts costly assessments to reduce outcome noise. Intensified assessment which shape the market’s inferences and provide implicit incentives for effort.

Society—including the evaluator and the market—holds dogmatic, inaccurate beliefs about ability, which distort learning about effort productivity and effort choice. This, in turn, shapes the evaluator’s assessment choice. We describe a feedback loop linking misspecified learning and distorted behaviors. We examine locally asymptotically stable steady states and link their robust comparative statics and learning foundations. Applications to education and labor market reveal how stereotypes can reinforce across domains—sometimes disguised as narrowing or even reversals of outcome gaps—and how policy interventions that intensify assessment may help.

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1 Introduction

We study a model of implicit incentive provision with misspecified learning, featuring an agent and an evaluator as strategic players and a market as a passive actor. An observable outcome depends on the agent’s innate ability, effort choice—whose effectiveness is governed by a productivity parameter—and noise. Based on the outcome, the market rewards the agent for their perceived ability and possibly their effort. The evaluator conducts costly assessments that reduce outcome noise, thereby shaping the market’s inferences and ultimately the agent’s effort incentives. All players are Bayesian, but the evaluator and the market—collectively referred to as society—hold dogmatic, inaccurate prior beliefs about the agent’s innate ability, while learning about their effort productivity from observed outcomes.

We shall first describe two situations that illustrate our framework and ideas.

Two students, Alice and Bob, are identical except for gender. Their school performance depends on innate ability and study effort, whose effectiveness reflects their study aptitude. The school acts as an evaluator, conducting costly assessments that reduce performance noise. Ultimately, both students are rewarded by the market based on their observed performance. Now, despite being identical, both the school and the market believe that Alice’s ability is lower than Bob’s. This reflects the *gender-brilliance stereotype*, which posits that girls lack the raw talent emphasized in certain fields such as STEM (Leslie et al., 2015; Bian et al., 2017). In reality, Alice and Bob are equally capable, so Alice’s strong performance comes as a surprise. In our framework, society does not update its belief about ability; instead, it misattributes Alice’s success to a greater aptitude for studying.

In the case where the school values effort *per se* (e.g., because higher student performance enhances its reputation or resources, or because it cares directly about the student’s outcome), the school scrutinizes Alice more intensively to induce greater effort. This raises Alice’s performance and shapes society’s subsequent learning about her study aptitude.

The second situation involves two entry-level workers, Chen and David, who are identical except for race and ethnicity. Job performance reflects innate ability but can be enhanced through costly effort, whose effectiveness captures work aptitude. Performance influences the market’s perception of ability and therefore the worker’s future career prospects. A manager conducts costly performance appraisals to reduce noise and make performance a more informative signal of ability. Society—including the manager and the market—believes that Chen is less capable than David, reflecting the *model minority stereotype* that portrays Asians as lacking creativity, leadership, and social skills (Lu, 2024).¹ As with Alice, Chen’s

¹Other biases and stereotypes may also contribute to an underestimation of minorities’ intrinsic abilities. The *ultimate attribution error* refers to the tendency to attribute out-group successes to external factors such as effort, and their failures to internal traits such as low ability (Pettigrew, 1979). The *stereotype incongruity*

strong performance is misattributed to greater work aptitude. The manager scrutinizes Chen more harshly, thereby distorting society’s subsequent learning about his work aptitude.

Our model captures the situations described above and more. We formalize the interaction among assessment, effort, and misspecified learning about productivity. We examine equilibrium beliefs and behavior, linking their comparative statics and learning foundations. In areas such as education and labor, our results identify new sources of discrimination, their welfare consequences, and policy levers that can curb discrimination.

Our analysis builds on the single-period version of Holmström’s (1999) career-concerns model, in which an agent’s outcome depends on random innate ability, effort, and noise. In our model, however, noise is reduced through the costly assessments conducted by an evaluator, who, like the market, may value effort in addition to ability. Intensified assessment increases the precision of the market’s inference about ability and strengthens the agent’s effort incentives. Moreover, we model the productivity of effort, which, together with mean ability, parameterize the mean outcome. The agent knows both parameters, whereas society must learn about them. Learning is Bayesian but begins from a misspecified prior. Along the ability dimension, the prior is degenerate and inaccurate, capturing the dogmatic stereotypes described above. Along the productivity dimension, the prior is nondegenerate and contains the true productivity in its support.

In a *Berk–Nash* equilibrium (Esponda and Pouzo, 2016), all players choose optimally given their beliefs, and beliefs are formed via Bayesian updating from the observed *outcome distribution* induced by equilibrium behaviors. Equilibrium beliefs concentrate on *KL-minimizers*, i.e., elements of the prior’s support that minimize the Kullback–Leibler divergence between the perceived and true outcome distributions. Ability misspecification therefore distorts equilibrium beliefs about productivity away from the truth. With moral hazard, this error prompts society to draw inaccurate inferences about effort choice—a model feature with novel consequences.

We study environments in which the agent works harder when more productive or when subject to more intensive assessment. We derive the perceived contribution of effort to the mean outcome, as well as the evaluator’s assessment strategy. Both are assumed to increase with society’s belief about productivity, and the former exhibits strict increasing differences in assessment and belief. Theorem 1 analyzes the direction of equilibrium distortions.

Two scenarios arise. In the case where society underestimates the agent’s ability yet observes better-than-expected outcomes, it reconciles the discrepancy by inferring that effort productivity is higher than it truly is. This overestimation leads to more intensive assess-

theory posits that when minority performance contradicts prevailing group stereotypes (e.g., East Asians excelling in leadership or vision), it is attributed to effort rather than innate talent (Eagly and Karau, 2002).

ment and raises the agent's effort choice and outcome, which in turn reduces the degree of overestimation needed to reconcile the initial discrepancy. Assessment and belief thus act as substitutes, yielding a unique equilibrium. By contrast, when society overestimates ability, it attributes the lower-than-expected outcome to productivity being lower than it truly is. Such underestimation lowers assessment, which triggers even more pessimistic beliefs. Multiple equilibria may arise, and some may exhibit large distortions even when ability misspecification is small; see Panel (a) of Figure 1.

Because multiple equilibria may arise and do not overall exhibit monotone comparative statics, we select a subset of equilibria based on their asymptotic stability. In Panel (a) of Figure 1, $\tilde{\psi}$ maps each degenerate productivity belief to the KL-minimizing posterior, formed under the optimal behavior given the initial belief. Berk-Nash equilibrium beliefs are the fixed point of $\tilde{\psi}$. An equilibrium is (*Lyapunov*) *locally asymptotically stable* (*resp. unstable*) if nearby beliefs are attracted to (*resp. repelled from*) it under $\tilde{\psi}$. With single-dimensional productivity, stability of an equilibrium is determined by whether $\tilde{\psi}$ crosses the 45-degree line from above or below. Online Appendix C provides learning foundations: in a dynamic model where beliefs and behavior co-evolve over time, society's long-run posterior almost surely concentrates on the set of stable-equilibrium beliefs, each with positive probability, while convergence to any unstable-equilibrium belief occurs with probability zero.

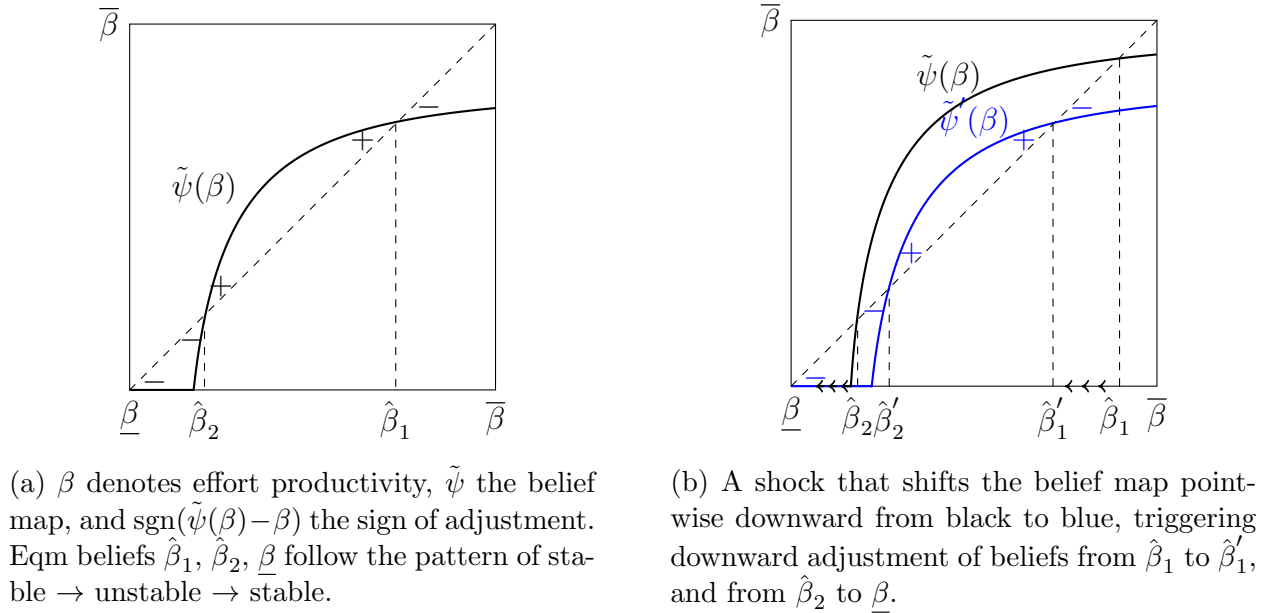


Figure 1: The case where society overestimates the agent's ability.

Theorem 2 derives monotone comparative statics for the set of stable equilibria. Panel (b) of Figure 1 illustrates the consequence of lowering assessment intensity for every degenerate productivity belief. When society overestimates the agent's ability, this shock shifts the map

$\tilde{\psi}$ pointwise downward to $\tilde{\psi}'$, because of the assessment-belief complementarity. Starting from a pre-shock equilibrium, the updated belief lies below its original level, triggering a downward adjustment. If the equilibrium is stable, as with $\hat{\beta}_1$, the adjustment settles at a nearby stable point $\hat{\beta}'_1$. If it is unstable, as with $\hat{\beta}_2$, the adjustment continues until the next stable point $\underline{\beta}$ is reached. Therefore, distortions associated with the set of stable-equilibrium beliefs increase in the weak set order. By contrast, comparative static of the unstable equilibrium, which predicts an upward adjustment $\hat{\beta}_2 \uparrow \hat{\beta}'_2$, lacks a learning foundation.

Our analysis yields insights into education and labor market discrimination. We discuss related empirical evidence in Section 3.2 and focus here on our conceptual contributions.

First, our result describes a cross-domain reinforcement of stereotypes. In equilibrium, society overestimates minority agents' effort productivity and imposes harsher scrutiny on them. In response, minorities exert greater effort than majorities. Yet the perceived effort gap is even larger because society uses inaccurate productivity beliefs to infer effort choices. Ability stereotypes thus spill over into the effort domain, leading to an overestimation of both minority agents' effort productivity and their effort choices.

Second, the welfare consequence of the *ability-effort tradeoff* described above hinge crucially on whether the market rewards effort. If only ability is rewarded—as in traditional career-concerns models—minorities earn less rewards despite exerting greater effort, resulting in lower objective welfare than majorities. If effort is also rewarded, then minorities are compensated for their higher perceived effort. The reward gap may therefore narrow or even reverse, although documented patterns consistent with this prediction may reflect an entrenchment of stereotypes across domains rather than genuine equity.

Third, intensifying assessment mitigates distortions in equilibrium beliefs and behavior. In education and labor contexts, assessment can be intensified directly by increasing the evaluator's valuation of effort, indirectly through raising the market's valuation of effort, or by reducing the cost of assessment or effort. We thus provide new perspectives on the anti-discrimination effects of a range of policies, including teacher pay-for-performance, curriculum alignment, general human capital investment, advances in assessment technology, and policies that intentionally or inadvertently alter effort costs.

Section 4 analyzes the role of affirmative action in a multi-agent variant of the model. The evaluator is required to apply the same assessment intensity to agents from both majority and minority groups, while still forming group-specific productivity beliefs based on group-specific outcomes. Such color-sighted affirmative action fundamentally alters the nature of the problem. In our single-agent baseline model, belief-behavior complementarity may trigger a cascade that drives equilibrium beliefs about majorities to the lower bound of the prior support; see Figure 1. Under affirmative action, this cascade is absent, because society

overestimates minorities’ productivity in equilibrium. Equal treatment requirement then imposes a tight universal lower bound on the optimal assessment intensity. Once this bound is reached, assessment wouldn’t further decrease, and the belief cascade would be halted.

Online Appendix C analyzes the misspecified Bayesian learning foundation of stable Berk–Nash equilibria. Under persistent asymmetric information about productivity and moral hazard, the agent’s effort choice is a random variable of their unknown productivity. The strategic uncertainty surrounding effort choice—which remains after society conditions on all available information—renders the posterior updating kernel nonparametric, even under linear–Gaussian primitives. We address this challenge by blending stochastic approximation with Bayesian statistics while accounting for players’ strategic behavior. Under affirmative action, the multidimensional belief space poses further challenges, which are addressed in Online Appendix D.

2 Model

2.1 Setup

Players, actions, and payoffs. Consider a game between three players: an evaluator (E), and agent (A), and the market (M). E and M are collectively referred to as *society* (S).

E moves first and commits to an assessment intensity $h \in [0, 1]$ of A. Assessment entails a cost $\kappa(h)$. A has a random innate ability $\eta \sim \mathcal{N}(\mu, 1)$ that is unobservable to all parties. After E’s move, A privately exerts an effort $a \geq 0$ at cost $c(a)$. Ability, effort, and assessment generate an outcome:

$$X = \underbrace{\eta}_{\text{innate ability}} + \underbrace{r(a, \beta)}_{\text{effective effort}} + \underbrace{\varepsilon}_{\text{noise}},$$

which reflects (1) ability, (2) the effect of effort $r(a, \beta)$, parameterized by A’s *effort productivity* $\beta \in [\underline{\beta}, \bar{\beta}] \subset \mathbb{R}_{++}$, and (3) a Gaussian error $\varepsilon \sim \mathcal{N}(0, h^{-1} - 1)$ whose variance decreases with the assessment intensity h . The outcome is publicly realized and observed by M. Based on the outcome, M forms an expectation about A’s market value and rewards A accordingly.

M values A at $\eta + v_M(a, \beta)$, where $v_M(a, \beta)$ denotes M’s valuation of effort and productivity. Traditional career-concerns models following Holmström (1999) have focused on the special case of $v_M \equiv 0$, in which only ability is rewarded but effort is not. Our framework nests such models but also allows $v_M \not\equiv 0$.

A’s payoff is M’s reward net of effort cost. E values effort and productivity at $v_E(a, \beta)$. E may or may not value ability.

We defer the assumptions on $\kappa(\cdot)$, $c(\cdot)$ and $v_E(\cdot, \cdot)$ to Appendix A, and proceed to present

two applications of our framework.

Application 1 (Education). A represents a student and E represents a school. The student's performance in school is $X = \eta + \beta a + \varepsilon$, which increases with both innate ability η and study effort a . β parameterizes study aptitude, and $r(a, \beta) = \beta a$ is the study output. The cost of studying is $c(a) = ca^2/2$.

The market rewards innate ability and a share $\delta \in [0, 1]$ of the study output, i.e. $v_M(a, \beta) = \delta \beta a$. A higher δ may result from reforms that better align the school's curricula with the market's needs.

The school benefits from high student performance (which enhances its reputation and resources) and possibly from high student well-being, assigning weights $\lambda_E > 0$ and $\lambda_A \geq 0$ to these considerations. Straightforward algebra shows that²

$$v_E(a, \beta) = \lambda_1 \beta a - \lambda_2 c(a) + \text{terms independent of effort},$$

$$\text{where } \lambda_1 = \lambda_E + \delta \lambda_A \quad \text{and} \quad \lambda_2 = \lambda_A.$$

The school can conduct assessment to reduce performance noise at cost $\kappa(h) = \kappa h^2/2$.

Application 2 (Labor). In the setting of Application 1, reinterpret E as an employer and A as a worker. X represents the worker's performance on the job, with a contribution of effort $r(a, \beta) = \beta a$ parameterized by work aptitude β . Noise in performance can be reduced by the employer via costly worker assessment.

The market values the worker at $\eta + \delta \beta a$. The case $\delta = 0$ corresponds to situations where effort generates only employer-specific human capital. By contrast, $\delta > 0$ if effort produces general human capital valued by the broader market.

The employer values the worker's performance (which corresponds to revenue), and it may additionally value the worker's well-being, with weights $\lambda_E > 0$ and $\lambda_A \geq 0$.

Priors. For $h > 0$, we may rewrite the outcome as

$$X = \mu + r(a, \beta) + \frac{\epsilon}{\sqrt{h}}, \quad \text{where } \epsilon := \sqrt{h}(\eta - \mu + \varepsilon) \sim \mathcal{N}(0, 1),$$

²Fix S's belief about the fundamental and A's effort choice as degenerate at (μ, β, a) . A standard calculation yields M's reward for A given X as

$$\mathbb{E}[\eta + \delta r(a, \beta) \mid X] = \mu + h[X - \mu - r(a, \beta)] + \delta r(a, \beta).$$

From E's perspective, X has expected value $\mu + r(a, \beta)$; therefore, A's expected payoff, after subtracting $c(a)$ from the right-hand side, equals $\mu + \delta r(a, \beta) - c(a)$. Taking the weighted sum of these objects by λ_E and λ_A , respectively, gives the expression above.

which makes it clear that $X \sim \mathcal{N}(\mu + r(a, \beta), h^{-1})$. The mean outcome $\mu + r(a, \beta)$ is parameterized by $\omega := (\mu, \beta)$, hereafter referred to as the *fundamental*. The true fundamental is $\omega^* = (\mu^*, \beta^*)$.

Players are Bayesian. At the outset, the true fundamental ω^* is revealed to A. That A knows ω^* is common knowledge. In our applications, this assumption reflects a shared belief that A is familiar with their own average talent and their effectiveness in studying or working. These are not accessible to S—E and M—who remain uncertain about the fundamental. Such uncertainty is represented by a prior over ω with support $\Omega = \{\hat{\mu}\} \times [\underline{\beta}, \bar{\beta}]$.

We assume that $\hat{\mu} \neq \mu^*$, meaning that S holds a dogmatic and inaccurate belief about A's mean ability. In our applications, this *misspecification* reflects stereotypes about minority groups' raw talents that are resistant to corrective information or interventions. Examples of such stereotypes were reviewed in Section 1.

The prior over effort productivity is supported on the compact interval $[\underline{\beta}, \bar{\beta}]$ satisfying $0 < \underline{\beta} < \beta^* < \bar{\beta}$. The assumption $\beta^* \in (\underline{\beta}, \bar{\beta})$ allows us to study how misspecification of ability distorts learning about effort productivity, even when the true productivity is considered a priori possible. The restriction $\underline{\beta} > 0$ has technical implications that will soon become clear.

A may or may not know S's prior over ω . This belief has no impact on A's strategic behavior.

Event sequence. Given players' priors, the game evolves as follows:

1. E publicly commits to an assessment intensity h .
2. A privately exerts an effort a .
3. Ability $\eta \sim \mathcal{N}(\mu^*, 1)$ and noise $\varepsilon \sim \mathcal{N}(0, h^{-1} - 1)$ are realized independently and determine the outcome as

$$X = \eta + r(a, \beta^*) + \varepsilon.$$

Only the realization of X (but not of η or ε) becomes public.

4. M forms an expectation about $\eta + v_M(a, \beta)$ and rewards A accordingly.

2.2 Equilibrium concept

Our solution concept is *Berk–Nash equilibrium*, which comprises behavior and beliefs such that (1) behavior is optimal given beliefs, and (2) beliefs are formed through Bayesian updating from the *outcome distribution* induced by behavior (Esponda and Pouzo, 2016).

In this section, we first describe these components and impose regularity conditions, and then formally define equilibrium.

Optimal effort. Fix an assessment intensity $h \in [0, 1]$, and suppose that S's belief about the fundamental and A's effort choice is degenerate at $(\omega, \hat{a}) \in \Omega \times \mathbb{R}_+$. Then X is believed to follow $\mathcal{N}(\hat{\mu} + r(\hat{a}, \beta), h^{-1})$. A standard calculation as in [Holmström \(1999\)](#), gives M's expectation of A's market value conditional on X :

$$\begin{aligned} \mathbb{E}_{\delta_{(\omega, \hat{a})}}[\eta + v_M(a, \beta) \mid X] &= \hat{\mu} + \frac{\text{Var}(\eta)}{\text{Var}(X)} [X - \hat{\mu} - r(\hat{a}, \beta)] + v_M(\hat{a}, \beta) \\ &= hX + \text{terms independent of actual effort.} \end{aligned}$$

On the right-hand side, only the mean of X is affected by the actual choice of effort. h is the precision of the outcome as a signal of innate ability, chosen by E to influence M's inference and, ultimately, to provide effort incentives to A.

Specifically, an agent with productivity β facing assessment h solves

$$\max_{a \geq 0} hr(a, \beta) - c(a). \quad (1)$$

Denote the solution to (1) by $a(h, \beta)$. The map $a : [0, 1] \times [\underline{\beta}, \bar{\beta}] \rightarrow \mathbb{R}_+$, $(h, \beta) \mapsto a(h, \beta)$ is A's *effort strategy*. A's actual choice of effort is $a(h, \beta^*)$, while S, taking A's productivity to be β , believes that A chooses $a(h, \beta)$. When $\beta \neq \beta^*$, actual and perceived effort differ.

Optimal assessment. Fix a degenerate belief at $\omega = (\hat{\mu}, \beta)$ held by E. Anticipating that A chooses $a(h, \beta)$ given assessment h , E formulates his problem by backward induction as

$$\max_{h \in [0, 1]} V_E(h, \beta) - \kappa(h), \quad \text{where} \quad V_E(h, \beta) := v_E(a(h, \beta), \beta). \quad (2)$$

Denote the solution by $h(\beta)$. The map $h : [\underline{\beta}, \bar{\beta}] \rightarrow [0, 1]$, $\beta \mapsto h(\beta)$ is E's *assessment strategy*.

Posterior belief. Fix an assessment intensity $h \in [0, 1]$ and the corresponding true effort $a(h, \beta^*)$. These behaviors determine the true outcome distribution. Suppose that S observes infinitely many independent draws from this distribution and updates beliefs about the fundamental via Bayes' rule. By [Berk \(1966\)](#), limiting posterior beliefs must concentrate on the set of KL-minimizers, i.e., elements in the prior support whose corresponding outcome distributions minimize the Kullback-Leibler (KL) divergence from the true distribution.

In our setting, define

$$R(h, \beta) := r(a(h, \beta), \beta)$$

as the *optimal effective effort* of an agent with productivity β who faces assessment intensity h . Then the true outcome distribution is $\mathcal{N}(\mu^* + R(h, \beta^*), h^{-1})$, while the outcome distribution for a given $\omega \in \Omega$ is $\mathcal{N}(\hat{\mu} + R(h, \beta), h^{-1})$. The KL minimization problem is

$$\min_{\omega \in \Omega} \frac{1}{2h^{-1}} [\hat{\mu} + R(h, \beta) - (\mu^* + R(h, \beta^*))]^2,$$

where the objective is the KL divergence. Since $\Omega = \{\hat{\mu}\} \times [\underline{\beta}, \bar{\beta}]$, and $h > 0$ holds on the relevant domain, this problem reduces to

$$\min_{\beta \in [\underline{\beta}, \bar{\beta}]} |\Delta + R(h, \beta) - R(h, \beta^*)|, \quad \text{where } \Delta := \hat{\mu} - \mu^*. \quad (3)$$

We denote the solution by $\tilde{\beta}(h)$ to emphasize its dependence on h .

Equilibrium concepts. Intersecting the equilibrium components yields:

Definition 1. A Berk–Nash equilibrium (Berk–NE) is a tuple $(a(\cdot, \cdot), \hat{h}, \hat{\beta})$ such that:³

- For any h and β , $a(h, \beta)$ is the effort choice of an agent with productivity β facing assessment h . The true effort is $a(h, \beta^*)$.
- $\hat{h} = h(\hat{\beta})$: assessment is optimal given E 's belief about the fundamental and the implied belief about the effort choice.
- $\hat{\beta} = \tilde{\beta}(\hat{h})$: belief results from Bayesian updating on the outcome distribution induced by $\hat{h}$ and $a(\hat{h}, \beta^*)$.

If $\hat{\beta}$ perfectly fits the equilibrium outcome distribution, yielding zero KL-divergence, then the Berk–NE is a self-confirming equilibrium (SCE).

It will be shown that equilibrium beliefs must be degenerate. As a result, we refer to $\hat{\beta}$ as an equilibrium belief, although strictly speaking it is $\delta_{\hat{\beta}}$. We consider all Berk–NE's and clearly indicate whether an equilibrium is self-confirming or not.⁴

Regularity conditions. We assume that the optimal choices of effort and assessment are always unique. For analytical convenience, we impose the following assumption:

Assumption 1. The mappings $(h, \beta) \mapsto a(h, \beta)$, $(h, \beta) \mapsto R(h, \beta)$, and $\beta \mapsto h(\beta)$ admit smooth extensions to $\beta \geq 0$ that satisfy the following properties:

³Throughout, we use $\hat{\cdot}$ to indicate equilibrium objects, and $\tilde{\cdot}$ to denote the projection of an object to $[\underline{\beta}, \bar{\beta}]$, i.e., $\tilde{x} = x$ if $x \in [\underline{\beta}, \bar{\beta}]$, $\underline{\beta}$ if $x < \underline{\beta}$, and $\bar{\beta}$ if $x > \bar{\beta}$.

⁴The concept of SCE originated in Fudenberg and Levine (1993) and is related to the alternative solution concepts developed in Jehiel (2005) and Spiegel (2020). Recently, Ba (forthcoming) and Lanzani (2025) study the emergence of SCE in misspecified learning models with paradigm shifts.

- (i) $a(\cdot, \cdot)$ satisfies $a(0, \beta) \equiv a(h, 0) \equiv 0$ and is strictly increasing in h for all $\beta > 0$ and in β for all $h \in (0, 1]$.
- (ii) $R(\cdot, \cdot)$ satisfies the properties assumed of $a(\cdot, \cdot)$ in Part (i) and strict increasing differences in (h, β) .
- (iii) $h(\cdot)$ is strictly increasing, $h(0) = 0$, and $h(\beta) \in (0, 1)$ for all $\beta > 0$.

In Appendix A, we impose sufficient conditions on model primitives to ensure Assumption 1. In Applications 1 and 2, solving equilibrium components yields

$$a(h, \beta) = \frac{h\beta}{c}, \quad R(h, \beta) = \frac{h\beta^2}{c}, \quad h(\beta) = \frac{\lambda_1\beta^2}{\lambda_2\beta^2 + \kappa c},$$

so Assumption 1 is satisfied.

3 Analysis

3.1 Preliminaries

This section introduces the preliminary concepts and notation for the main analysis.

We develop a fixed-point representation of Berk–NE. Since $R(h, \beta)$ is strictly increasing in β for all $h > 0$, (3) admits a unique solution, $\tilde{\beta}(h)$, for any such h . This is in particular true over the range $h([\underline{\beta}, \bar{\beta}])$, as $h([\underline{\beta}, \bar{\beta}]) \subset \mathbb{R}_{++}$ under Assumption 1 (iii) and $\underline{\beta} > 0$.⁵

Define a function

$$\tilde{\psi} : [\underline{\beta}, \bar{\beta}] \rightarrow [\underline{\beta}, \bar{\beta}], \quad \beta \mapsto \tilde{\psi}(\beta) := (\tilde{\beta} \circ h)(\beta).$$

For each $\beta \in [\underline{\beta}, \bar{\beta}]$, $\tilde{\psi}(\beta)$ is the unique KL minimizer for the outcome distribution induced by the optimal assessment given belief β . The fixed points of $\tilde{\psi}$ are the Berk–NE beliefs about the productivity parameter. Moreover, if a fixed point $\hat{\beta}$ lies in $(\underline{\beta}, \bar{\beta})$, then it is a SCE belief. This is because $\hat{\beta}$ best fits the outcome distribution without hitting the boundaries, so the fit must be perfect. The existence of a Berk–NE follows from Brouwer’s fixed point theorem, while an SCE may not exist.

A main challenge faced in our analysis is equilibrium multiplicity. In Applications 1 and 2,

$$\tilde{\psi}(\beta) = \sqrt{\max \left\{ (\beta^*)^2 - \frac{\Delta c}{h(\beta)}, (\underline{\beta})^2 \right\}}.$$

⁵Since $R(0, \beta) = 0$ for all β , $\tilde{\beta}(h)$ admits the entire interval $[\underline{\beta}, \bar{\beta}]$ as solutions when $h = 0$. The assumption $\underline{\beta} > 0$ ensures that $\tilde{\beta}(h)$ is unique for all $h \in h([\underline{\beta}, \bar{\beta}])$.

Panel (a) of Figure 1 plots $\tilde{\psi}$ and its fixed points for a small, positive Δ . Multiple (interior) fixed points may arise and, as we will demonstrate, do not exhibit monotone comparative statics as we vary model primitives.

We deal with these challenges by using out-of-equilibrium dynamics to select asymptotically stable equilibria. The analysis of these dynamics is deferred to Online Appendix C. Here, we introduce the notion of stability and discuss its motivation.

To begin, we make the following assumption about $\tilde{\psi}$:

Assumption 2. *$\tilde{\psi}$ strictly crosses the 45 degree line at its fixed points.*

Assumption 2 states that all roots of the function $\tilde{\psi}(\beta) - \beta$ are regular, which holds generically by an application of Sard’s lemma; details are in Appendix A.

Under Assumption 2, we can classify Berk–NE into two categories:

Definition 2. *A Berk–NE is locally asymptotically stable in the Lyapunov sense if $\tilde{\psi}$ crosses the 45-degree line strictly from above at the corresponding $\hat{\beta}$. It is unstable if crossing occurs strictly from below.*

The concepts of stability and instability take on their natural meaning under a dynamic model of out-of-equilibrium learning with misspecification. In Online Appendix C, S begins with a misspecified prior and updates beliefs via Bayes’ rule based on observed outcomes over an infinite horizon. Players’ behavior evolves with their beliefs and, in turn, influences subsequent belief updating. This dependence between beliefs and behavior renders the standard law of large numbers inapplicable. Nevertheless, under mild regularity conditions, S’s posterior beliefs are shown to converge with probability one to the set of locally asymptotically stable values of $\hat{\beta}$, with each such $\hat{\beta}$ reached with positive probability. By contrast, convergence to any unstable value of $\hat{\beta}$ occurs with zero probability. We therefore adopt local asymptotic stability as the equilibrium selection criterion.

The core argument presented in Online Appendix C is that, asymptotically, S’s posterior beliefs become almost degenerate, and their evolution can be approximated by a deterministic ordinary differential equation (ODE). Along the relevant dimension, the drift of the ODE has the same sign as that of $\tilde{\psi}(\beta) - \beta$. A steady state of the ODE satisfies $\tilde{\psi}(\beta) = \beta$ and corresponds to a Berk–NE belief. Starting from a steady state, small perturbations trigger adjustments that restore beliefs to their original level if the steady state is locally asymptotically stable, and push beliefs further away if it is unstable. Panel (a) of Figure 1 illustrates these adjustment dynamics.

3.2 Main results

We present two main theorems. The first addresses equilibrium uniqueness, stability, and the direction of distortions. It is presented in two separate parts that deal with the cases $\Delta > 0$ and $\Delta < 0$.

Theorem 1 (Part A). *Suppose Assumptions 1 and 2 hold. When S underestimates A 's ability, i.e., $\Delta < 0$, there is a unique Berk-NE:*

- (i) *The equilibrium is locally asymptotically stable.*
- (ii) *S 's equilibrium belief about effort productivity is above the truth: $\hat{\beta} > \beta^*$.*
- (iii) *For $|\Delta|$ sufficiently small, the equilibrium is self-confirming with $|\hat{\beta} - \beta^*| = O(\Delta)$.⁶*

Theorem 1 (Part B). *When S overestimates A 's ability, i.e., $\Delta > 0$, there is a finite and odd number K of Berk-NE, with $\hat{\beta}_1 < \dots < \hat{\beta}_K$:*

- (i) *Equilibrium K is locally asymptotically stable, equilibrium $K - 1$ is unstable, and this alternating pattern continues.*
- (ii) *All equilibrium beliefs about effort productivity are below the truth: $\hat{\beta}_K < \beta^*$.*
- (iii) *All equilibria, except possibly one, are self-confirming. For any $\Delta > 0$, there exists $\check{\beta} > 0$ such that whenever $\underline{\beta} \in (0, \check{\beta})$, $\hat{\beta}_1 = \underline{\beta}$ and equilibrium 1 is not self-confirming.*

The second theorem derives monotone comparative statics for the set of belief distortions associated with *all* locally asymptotically stable equilibria.

Theorem 2. *Under Assumptions 1 and 2, the set of belief distortion $|\hat{\beta} - \beta^*|$'s in locally asymptotically stable equilibria increases in the weak set order when either:*

- (i) *The degree of ability misspecification $|\Delta|$ increases.*
- (ii) *The assessment intensity $h(\beta)$ decreases, and so incentives weaken for all productive types $\beta > 0$.*

Furthermore, the above changes yield distinct sets of SCE's (if nonempty).

We examine the implications of Theorems 1 and 2 for discrimination. Consider two groups of agents, m and w , who are identical in reality. S overestimates m 's innate ability and underestimates w 's, i.e., $\Delta_m > 0 > \Delta_w$, where Δ_i denotes the misspecification of group i 's ability.

The next proposition compares the equilibrium outcomes and welfare between m and w .

⁶We use the Landau notation: $O(\cdot)$ means at most on the order of, $\Theta(\cdot)$ means exactly on the order of, and $o(\cdot)$ means slower than the order of.

Proposition 1. *In Applications 1 and 2, the following are true in any SCE:*

- (i) *S underestimates m 's effort productivity and overestimates w 's: $\hat{\beta}_w > \beta^* > \hat{\beta}_m$.*
- (ii) *w faces more intensive assessment than m : $h(\hat{\beta}_w) > h(\hat{\beta}_m)$.*
- (iii) *While w 's true effort is greater than m 's, their perceived effort gap is even larger:*

$$a(h(\hat{\beta}_w), \hat{\beta}_w) > a(h(\hat{\beta}_w), \beta^*) > a(h(\hat{\beta}_m), \beta^*) > a(h(\hat{\beta}_m), \hat{\beta}_m).$$

- (iv) *The expected market reward to group i , evaluated under their true outcome distribution, equals $\hat{\mu}_i + \delta \hat{\beta}_i a(h(\hat{\beta}_i), \hat{\beta}_i)$. The difference between m and w equals*

$$\underbrace{\hat{\mu}_m - \hat{\mu}_w}_{>0} + \delta \underbrace{\left[\overset{(1)}{\hat{\beta}_m} \overset{(2)}{a(h(\hat{\beta}_m), \hat{\beta}_m)} \overset{(3)}{-\hat{\beta}_w} \overset{(1)}{a(h(\hat{\beta}_w), \hat{\beta}_w)} \overset{(2)}{\overset{(3)}}{\right]} \right]}_{<0}.$$

- (a) *When effort is not rewarded by the market, i.e., $\delta = 0$, m out-earns w .*
- (b) *When effort is rewarded by the market, i.e., $\delta > 0$, it is possible for w to earn more than m , but only because of (1) higher perceived productivity, (2) more intensive assessment, and (3) overestimation of effort based on inflated productivity belief. These channels are highlighted above.*
- (v) *The difference in the objective welfare between m and w —obtained by subtracting the true effort cost $c(a(h(\hat{\beta}_i), \beta^*))$ from the expected market reward in Part (iv)—is generally ambiguous. When $\delta = 0$, m is better-off than w .*

The next proposition identifies environmental shocks that mitigate the set of belief distortions in stable equilibria.

Proposition 2. *In Applications 1 and 2, the set of belief distortion $|\hat{\beta} - \beta^*|$'s in locally asymptotically stable equilibria decreases in the weak set order, when either E or M values effort more, or the cost of effort or assessment decreases. That is, any $\zeta \in \{\lambda_E, \delta, -c, -\kappa\}$ increases, so the optimal assessment $h(\beta, \zeta)$, parameterized by ζ , increases for all $\beta > 0$.*

3.3 Implications

Our results have far-reaching consequences for education and labor market discrimination. To streamline presentation, we focus on education discrimination in this section and postpone the discussion of labor market discrimination until Section 5.1.

Cross-domain reinforcement of stereotypes. In education, girls are routinely cast as less brilliant than boys in STEM, where “raw talent” is framed as the key to success (Leslie et al., 2015; Bian et al., 2017). Parents, teachers, and broader society often assume that girls can only keep up by working harder, crediting their achievements to effort and study skills rather than innate ability (Wang and Hall, 2018). This bias, known in educational psychology as the *talent–effort trade-off*, exposes girls to harsher scrutiny. Tiedemann (2000) and Copur-Gencturk et al. (2020) show that teachers give more critical feedback on girls’ mathematics work, calling out minor mistakes that would be ignored in boys’ assignments. This heightened pressure fuels socially prescribed perfectionism and burnout among girls (Rice et al., 2015), suggesting that girls work harder than boys in practice.

A similar pattern plays out along racial lines: Asian students are routinely cast as hard-working but uncreative (Lu, 2024). In college admissions, Asian applicants are systematically assigned lower “personal” ratings that are supposed to reflect leadership, charisma, and social skills (Arcidiacono et al., 2022). Backed into this corner, Asian parents pour money into “Ivy coaching,” hiring private counselors to polish their children’s packages (Moscatello, 2024). Their children, in turn, are pushed to “out-white” their white peers by working even harder and jumping through more hoops (Hsin and Xie, 2014).

These findings align with Parts (i)–(iii) of Proposition 1 on perceived effort productivity, assessment, and real effort. Part (iii) paints an even starker picture: society ultimately comes to see minority students as exerting more effort than they actually do. This misperception is not driven by the fact that minorities are more heavily scrutinized and do in fact work harder. Instead, it is fueled by inflated beliefs about their effort productivity, which lead to overestimation of their effort choices. Together, Proposition 1 predicts a cross-domain amplification of stereotypes: stereotypes about ability spills over into the effort domain, warping beliefs about both effort productivity and choice.

Reward and welfare. Parts (iv) and (v) of Proposition 1 highlight the welfare consequences of stereotypes, emphasizing the role of market’s reward for effort.

When the market rewards ability alone, as in traditional career-concerns models (Holmström, 1999), majorities receive higher rewards due to their greater perceived ability while exerting less effort, therefore attaining higher objective welfare than minorities.

When effort is valued by the market, e.g., through curriculum alignment, minorities may out-earn majorities, consistent with narrowing or reversed gender gaps in STEM and strong admissions outcomes for Asian students (Ceci et al., 2014; Lau, 2024). Part (iv) of Proposition 1 cautions that such patterns need not indicate genuine equity. Instead, they may reflect entrenched ability stereotypes that spill into effort, manifested in overestimated effort productivity and choice, and in greater scrutiny, with ambiguous welfare consequences.

Distortion mitigation via intensifying assessment. Part (ii) of Proposition 2 identifies distortion-mitigating interventions that operate through intensifying assessment.

In education, λ_E and δ parameterize the school’s valuation of effort and the market’s reward to effort, respectively. To the extent that the school values student well-being, a higher δ indirectly raises its valuation of effort. Policy interventions can increase λ_E through pay-for-performance for teachers (Deming, 2022) and increase δ through curriculum alignment (Cullen et al., 2013). Both interventions intensify the school’s assessment of students.

Meanwhile, c and κ parameterize the costs of effort and assessment, respectively. A reduction in either intensifies the school’s assessment. In practice, data-driven assessment innovations can reduce κ (Mandinach and Schildkamp, 2021), whereas several policies raise c , sometimes inadvertently among students from less affluent backgrounds. These include China’s ban on after-school tutoring (Hall and Chen, 2024), the availability of multiple standardized-test retakes in the United States (Proulx, 2024), and college-admissions practices that favor affluent applicants through legacy preferences, non-academic credentials, and athletic recruitment (Chetty et al., 2023).

3.4 Proofs

We outline the proofs of Theorems 1 and 2. See Appendix B for minor details and the proof of Proposition 1. Proposition 2 holds because E’s payoff, $V_E(h, \zeta)$, has strict increasing differences in (h, ζ) .

Proof of Theorem 1. Since $R(h, \beta)$ is strictly increasing in β for all $h > 0$, it follows that, for each $\beta \in [\underline{\beta}, \bar{\beta}]$, the following equation in x :

$$R(h(\beta), x) - R(h(\beta), \beta^*) = -\Delta, \quad x \in \mathbb{R}_+ \quad (4)$$

admits at most one root. In this section, we assume that a root exists and denote it by $\psi(\beta)$. Appendix B deals with situations in which a root may not exist.

By construction, $\psi(\beta)$ is the unconstrained best fit of the outcome distribution induced by the optimal assessment for β . Projecting $\psi(\beta)$ onto $[\underline{\beta}, \bar{\beta}]$ yields the constrained best fit $\tilde{\psi}(\beta)$. A Berk–NE with $\hat{\beta}$ is self-confirming if $\psi(\hat{\beta}) = \hat{\beta}$.

The remainder of the proof distinguishes two cases: $\Delta < 0$ and $\Delta > 0$.

Case A. When $\Delta < 0$, the right-hand side of (4) is positive. Consequently, $\psi(\beta) > \beta^*$ holds for all $\beta \in [\underline{\beta}, \bar{\beta}]$. As we raise β , $h(\beta)$ increases. Since $R(h, \beta)$ has strictly increasing differences, the left-hand side of (4) increases for all fixed $x > \beta^*$. Restoring equality

therefore requires that $\psi(\beta)$ decreases with β . After projecting, $\tilde{\psi}(\beta)$ remains decreasing, strictly so whenever $\tilde{\psi}(\beta) < \bar{\beta}$.

As illustrated in Panel (a) of Figure 2, $\tilde{\psi}$ has a unique fixed point, which corresponds to a locally asymptotically stable Berk–NE. For sufficiently small $|\Delta|$, we have $|\hat{\beta} - \beta^*| = O(|\Delta|)$, so the equilibrium is self-confirming.

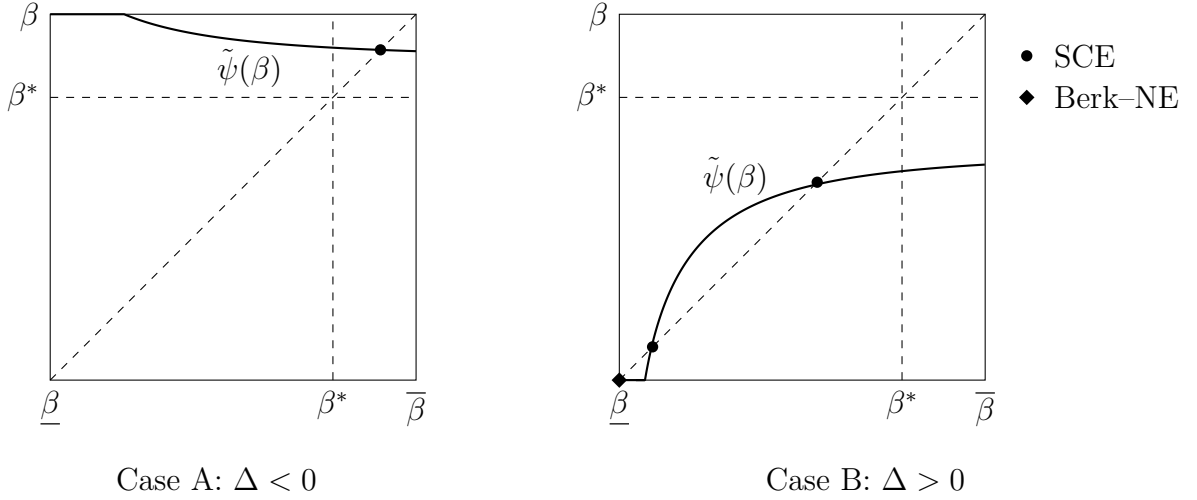


Figure 2: Equilibrium uniqueness, stability, and distortion.

Case B. When $\Delta > 0$, the argument is analogous to Case A, with the relevant signs reversed. We therefore conclude that $\tilde{\psi}(\beta)$ is strictly bounded above by β^* and is increasing in β , strictly so whenever $\tilde{\psi}(\beta) > \underline{\beta}$. As depicted in Panel (b) of Figure 2, multiple fixed points may arise, among which the largest is locally asymptotically stable, the second largest is unstable, and so on. All fixed points—except possibly the smallest—lie in $(\underline{\beta}, \bar{\beta})$ and correspond to SCE beliefs.

When the smallest fixed point equals $\underline{\beta}$, the corresponding equilibrium is not an SCE. We now demonstrate that this situation arises for every $\Delta > 0$, provided that $\underline{\beta}$ is sufficiently close to zero. Recall the assumptions that an unproductive agent shouldn't be assessed, and that an unassessed agent exerts no effort, i.e., $h(\beta) = 0$ and $R(0, \beta) = 0$. Therefore, for every $\Delta > 0$, there exists $\check{\beta} \in (0, \beta^*)$ such that

$$R(h(\beta), \beta) - R(h(\beta), \beta^*) > -\Delta \quad \forall \beta \in [0, \check{\beta}).$$

Pick any $\underline{\beta} \in (0, \check{\beta})$. Since $\psi(\underline{\beta})$ solves (4), we must have $\psi(\underline{\beta}) < \underline{\beta}$ and so $\tilde{\psi}(\underline{\beta}) = \underline{\beta}$. Thus, $\underline{\beta}$ is a Berk–NE belief but not an SCE belief. $\square$

Proof of Theorem 2. When $\Delta > 0$, an increase in Δ makes the right-hand side of (4) more negative, requiring a smaller solution x . The same conclusion follows from a pointwise reduction in $h(\beta)$ for all $\beta > 0$: because $R(h, \beta)$ has strict increasing differences, the solution x again decreases. In either case, $\psi(\cdot)$ and hence $\tilde{\psi}(\cdot)$ shift downward pointwise, as illustrated in Panel (b) of Figure 1. The comparative statics of the stable-equilibrium set then follow from the correspondence principle, which also implies that the opposite movements of unstable equilibria lack learning foundations. The argument for $\Delta < 0$ is analogous. $\square$

Remark 1. Theorem 1 (iii) holds because assessment choice remains unconstrained as S varies its belief, i.e., $h([\underline{\beta}, \overline{\beta}]) \subset (0, 1)$. When $\underline{\beta} \in (0, \check{\beta})$, i.e., the belief space is not too constrained from below, assessment-belief complementarity triggers a cascade that drives the smallest equilibrium belief to $\underline{\beta}$. While natural in single-agent environments, these model features may no longer hold in multi-agent environments, a topic we now turn to.

4 Affirmative action

In this section, we consider a variant of the baseline model where E is required by affirmative action to apply a common assessment intensity across multiple groups of agents.

Setup. There are $J \geq 2$ groups of agents, each with population share $\alpha_j > 0$ satisfying $\sum_{j=1}^J \alpha_j = 1$. The true mean ability and effort productivity of group j are $\mu_j^* \in \mathbb{R}$ and $\beta_j^* \in (\underline{\beta}, \overline{\beta})$, respectively. The game begins with E committing to a common assessment intensity $h \in [0, 1]$, which applies to all groups as mandated by affirmative action policies. Each group j then privately chooses effort $a_j \in \mathbb{R}_+$. Group j 's outcome is

$$X_j = \eta_j + r(a_j, \beta_j^*) + \varepsilon_j,$$

where $\eta_j \sim \mathcal{N}(\mu_j^*, 1)$ and $\varepsilon_j \sim \mathcal{N}(0, h^{-1} - 1)$ are independent both of each other and across groups. All groups' outcomes are publicly observable, from which M infers each group's market value and determines rewards.

Let $\boldsymbol{\mu} := (\mu_j^*)_{j=1}^J$ and $\boldsymbol{\beta} := (\beta_j^*)_{j=1}^J$ denote the vectors of mean abilities and effort productivities, respectively. The vector $\boldsymbol{\omega} := (\boldsymbol{\mu}, \boldsymbol{\beta})$ denotes a fundamental. The true fundamental is $\boldsymbol{\omega}^*$. It is common knowledge that each group j knows its own mean ability and effort productivity, i.e., the j -th coordinate of $\boldsymbol{\omega}^*$. Group j may or may not know the remaining components of $\boldsymbol{\omega}^*$, and such beliefs are irrelevant for its strategic behavior.

S holds a prior over $\boldsymbol{\omega}$ that factorizes over groups, with support $\prod_{j=1}^J (\{\hat{\mu}_j\} \times [\underline{\beta}, \overline{\beta}])$. Along the ability dimension, we assume that $\hat{\mu}_j \neq \mu_j^*$ for each j and denote $\Delta_j := \hat{\mu}_j - \mu_j^*$.

Both sets $\{j : \Delta_j < 0\}$ and $\{j : \Delta_j > 0\}$ are taken to be nonempty, so S overestimates the abilities of some groups and underestimates those of others. Along the productivity dimension, the prior is supported on $[\underline{\beta}, \bar{\beta}]^J$, which contains β^* in its interior.

The affirmative action policy of interest is *color-sighted*, in that E still forms group-specific beliefs about productivity based on group-specific outcomes.⁷ The equal-treatment restriction then renders E's problem non-separable across groups and multidimensional in nature, requiring new concepts and methods to analyze.

Concepts. Given an assessment intensity $h \in [0, 1]$, each group j faces the same optimal effort-choice problem as previously. E's problem is different. Given $\beta \in [\underline{\beta}, \bar{\beta}]^J$, the optimal assessment intensity solves

$$\max_{h \in [0, 1]} \sum_{j=1}^J \alpha_j V_E(h, \beta_j) - \kappa(h). \quad (5)$$

Under the same conditions that underpin Assumption 1, this problem admits a unique solution, denoted by $h(\beta)$. The assessment strategy is $h : [\underline{\beta}, \bar{\beta}]^J \rightarrow [0, 1]$.

Given h , the vector of true outcomes follows a J -dimensional Gaussian distribution with mean vector $(\mu_j^* + R(h, \beta_j^*))_{j=1}^J$ and covariance matrix $h^{-1} Id_J$. From S's perspective, the mean vector is, instead, $(\hat{\mu}_j + R(h, \beta_j))_{j=1}^J$. The KL minimization problem is

$$\min_{\beta \in [\underline{\beta}, \bar{\beta}]^J} \frac{1}{2h^{-1}} \sum_{j=1}^J [\Delta_j + R(h, \beta_j) - R(h, \beta_j^*)]^2. \quad (6)$$

where the objective is the KL divergence. Because this problem is separable across groups, its minimizer is $\tilde{\beta}(h) := (\tilde{\beta}_j(h))_{j=1}^J$.

Berk-NE beliefs about β are the fixed points of

$$\tilde{\psi} : [\underline{\beta}, \bar{\beta}]^J \rightarrow [\underline{\beta}, \bar{\beta}]^J, \quad \beta \mapsto \tilde{\psi}(\beta) := (\tilde{\beta} \circ h)(\beta).$$

Fixed points in the interior of $[\underline{\beta}, \bar{\beta}]^J$ correspond to SCE beliefs.

A Berk-Nash NE with belief vector $\hat{\beta}$ is locally asymptotically stable in the Lyapunov sense if all eigenvalues of the Jacobian of $\tilde{\psi}(\beta) - \beta$, evaluated at $\hat{\beta}$, have negative real parts.

Result. We focus on situations in which each group's ability is slightly misspecified to draw the sharpest distinction between the baseline model.

Theorem 3. *The following are true when $|\Delta_j|$'s are sufficiently small:*

⁷By contrast, *blind assessment* only reveals the aggregate outcome to E. In our setting, this leads to a lack of identification, as KL minimization requires choosing a J -dimensional vector β to minimize the scalar difference $|\sum_{j=1}^J \alpha_j [\mu_j^* + R(h, \beta_j^*)] - \sum_{j=1}^J \alpha_j [\hat{\mu}_j + R(h, \beta_j)]|$ between the true and perceived aggregate means.

- (i) *There is a unique Berk-NE, which is self-confirming and locally asymptotically stable.*
- (ii) *In this equilibrium, $|\hat{\beta}_j - \beta_j^*| = O(|\Delta_j|)$, with $\hat{\beta}_j < \beta_j^*$ if $\Delta_j > 0$ and $\hat{\beta}_j > \beta_j^*$ if $\Delta_j < 0$.*
- (iii) *A small increase in $|\Delta_j|$ raises $|\hat{\beta}_j - \beta_j^*|$ but has a negligible impact on $|\hat{\beta}_k - \beta_k^*|$ for $k \neq j$.*
- (iv) *A small increase in ζ raises all $|\hat{\beta}_j - \beta_j^*|$, $j = 1, \dots, J$.*

To illustrate the difference between Theorems 1 and 3, consider, in the context of Application 1, two groups m and w with $\Delta_w < 0 < \Delta_m$. When m is assessed separately from w , S underestimates β_m , and the distortion from β_m^* can be large even for a small $\Delta_m > 0$, in the sense formalized in Theorem 1. As noted in Remark 1, this result hinges on the assumption that assessment choice remains unconstrained as S lowers its belief about β_m ; recall that

$$h(\beta) = \frac{\lambda_1 \beta^2}{\lambda_2 \beta^2 + \kappa c}, \quad \text{with } h([\underline{\beta}, \bar{\beta}]) \subset (0, 1).$$

As assessment declines, beliefs fall, triggering further downward revisions of assessment. The resulting cascade can drive equilibrium beliefs down to $\underline{\beta}$.

Conversely, Remark 1 reveals an effective way to halt the cascade: imposing a lower bound on assessment intensity. By Theorem 3, such a bound arises as a by-product of color-sighted, equal assessment. The reason is that, since w constitutes a fraction $\alpha_w > 0$ of the population and S must overestimate their productivity in equilibrium, the optimal assessment intensity is bounded universally below by

$$h(\beta) = \frac{\lambda_1 \sum_{j \in \{m, w\}} \alpha_j \beta_j^2}{\lambda_2 \sum_{j \in \{m, w\}} \alpha_j \beta_j^2 + \kappa c} > \check{h} := \frac{\lambda_1 \alpha_w (\beta_w^*)^2}{\lambda_2 \alpha_w (\beta_w^*)^2 + \kappa c}.$$

Universality refers to the property that $\check{h}$ depends only on w 's true characteristics, for reasons stated above.

Parts (iii) and (iv) of Theorem 3 analyze the comparative statics of equilibrium belief distortions under affirmative action. Results are qualitatively similar to those in the baseline model, although the proofs are more evolved; see Appendix B.

Agent welfare can be compared using the formulae in Proposition 1. No clear-cut conclusion can be drawn in general, consistent with Part (v) of the proposition.

In Online Appendix D, we study the misspecified Bayesian learning foundation of the Berk-NE under affirmative action. Because β is multidimensional, the invariant set of the limiting ODE may contain complex objects—such cycles and strange attractors—in addition to the steady state $\hat{\beta}$. Nevertheless, we show that the Jacobian of the limiting

ODE has eigenvalues with all negative real parts near $\hat{\beta}$, so $\hat{\beta}$ is a sink and hence a point attractor. Applying the “attractor theorem” of [Benaïm \(1999\)](#), we show that posterior beliefs converge to $\hat{\beta}$ with positive probability, and with arbitrarily high probability when the prior is sufficiently concentrated around $\hat{\beta}$.

5 Discussions

5.1 Lessons for labor market discrimination

Given space constraints, we focus on gender discrimination in the labor market and on insights not immediate from education discrimination. See, however, [Hyun \(2005\)](#) and [Lu \(2024\)](#) on how the model-minority stereotype impedes East Asians’ career advancement, a phenomenon known as the *bamboo ceiling*.

Our prediction that women face heightened scrutiny and exert greater effort is supported by evidence from academia. For instance, [Correll \(2017\)](#) find that professors evaluating fictitious candidates write more doubt-raising comments for women than for men, such as “I would need to see evidence that she had gotten these grants and publications on her own.” [Hengel \(2022\)](#) finds that female-authored papers take longer to pass peer review than comparable male-authored papers, and that female authors invest more in writing quality. [Eberhardt et al. \(2023\)](#) find that recommendation letters for entry-level faculty positions are more likely to describe female applicants using “grindstone” terms (e.g., hardworking, diligent) and make more frequent comparisons to their peers. [Dupas et al. \(2021\)](#) document that female speakers receive more probing seminar questions.

When workers’ effort constitutes general human-capital investment valued by the broader market, $\delta > 0$. Proposition 1 predicts a narrowing or even reversal of the gender earnings gap—consistent with the erosion of the glass ceiling documented by [Gayle et al. \(2012\)](#) and [Bertrand \(2018\)](#)—yet cautions against interpreting such trends as definitive evidence of equity, as they may instead reflect the cross-domain spillover of stereotypes.

Conventional anti-stereotype interventions, such as longitudinal debiasing training and structured intergroup contact, are typically studied and deployed in isolation ([Devine et al., 2012](#)). These programs are now widespread but costly to scale, underscoring the need for leaner tools ([Bohnet, 2016](#)). Part (i) of Theorem 2 identifies a potentially cost-saving lever: partially correcting misspecified ability beliefs also attenuates distortions in productivity beliefs. Cross-domain reinforcement can thus be harnessed to correct stereotypes along multiple margins simultaneously.

Finally, assessment can be intensified—and equilibrium distortions mitigated—through

general human-capital subsidies, which raise δ ; rater training, which lowers assessment costs κ (Bohnet, 2016); and AI–human complementarities, which reduce effort costs c (Brynjolfsson et al., 2025).

5.2 Strategic uncertainty

A key feature of our model is that S uses its belief about A’s productivity to infer effort choice. Inaccurate productivity beliefs distort effort inferences, even though the equilibrium effort strategy $a(\cdot, \cdot)$ is known. We refer to this model feature as *strategic uncertainty*; the rationale for this terminology will soon become clear.

In Section 2.1, we already discussed why strategic uncertainty arises naturally in our applications featuring persistent asymmetric information about productivity and moral hazard. We now examine its equilibrium and learning consequences by comparing our model with those in the literature without it.

Heterogeneous priors. The case where multiple players hold heterogeneous, common-knowledge priors about the fundamental has been examined by Ba and Gindin (2023) and Murooka and Yamamoto (2021, 2023) among others. In our setting, suppose it is common knowledge that S’s prior about ω is supported on $\{\hat{\mu}\} \times [\underline{\beta}, \bar{\beta}]$, while A’s prior is degenerate at (μ^*, β^*) . The analysis remains unchanged if A’s prior is supported on $\{\mu^*\} \times [\underline{\beta}, \bar{\beta}]$; that is, A knows his true mean ability but must learn about his productivity.

Under this alternative framework, A’s true effort choice: $a(h, \beta^*)$, is commonly known. For a given assessment intensity h , S perceives the effective effort as

$$R^F(h, \beta) := r(a(h, \beta^*), \beta),$$

where productivity belief enters only through the fundamental channel, not through perceived effort choice. The KL minimization problem faced by S becomes

$$\min_{\beta \in [\underline{\beta}, \bar{\beta}]} |\Delta + R^F(h, \beta) - R^F(h, \beta^*)|.$$

One can verify that Theorems 1 and 2 remain qualitatively valid under this alternative framework; details are available upon request. The next proposition compares the magnitudes of equilibrium belief distortions when ability misspecification is small. Relative to our model, heterogeneous priors amplify the belief distortion associated with the least-distorted SCE: the one with the smallest $\hat{\beta}$ when $\Delta < 0$, and the one with the largest $\hat{\beta}$ when $\Delta > 0$.

Proposition 3. *When $|\Delta|$ is sufficiently small, the belief distortion $|\hat{\beta} - \beta^*|$ in the least distorted SCE is smaller in our model than under heterogeneous priors.*

To develop intuitions for Proposition 3, we decompose $R(h, \beta) - R(h, \beta^*)$ as

$$R(h, \beta) - R(h, \beta^*) = \underbrace{r(a(h, \beta), \beta) - r(a(h, \beta^*), \beta)}_{(I)} + \underbrace{r(a(h, \beta^*), \beta) - r(a(h, \beta^*), \beta^*)}_{(II)=R^F(h, \beta) - R^F(h, \beta^*)}.$$

On the right-hand side, (I) and (II) capture the effects of misperceiving effort choice and productivity alone, respectively. The total effect is on the left-hand side.

Suppose that $\Delta > 0$. Then equilibrium beliefs about productivity must be downward distorted relative to the truth. Since (I) is negative over the relevant domain $[\underline{\beta}, \beta^*)$, adding it to (II) lowers the left-hand side of the SCE-defining equation under heterogeneous priors: $R^F(h(\beta), x) - R^F(h(\beta), \beta^*) = -\Delta$. Restoring equality requires increasing x , which mitigates equilibrium belief distortions. The argument for $\Delta < 0$ is analogous.

The above argument holds E's assessment strategies fixed across scenarios. In Appendix B, we show that, when ability misspecification is small, allowing the assessment strategy to vary has only a second-order effect on the least distorted SCE.

The comparisons between other equilibrium objects are in general ambiguous. Consider, for instance, assessment intensity. Let δ_x denote S's first-order belief about productivity, and let δ_{δ_y} denote its second-order belief about A's first-order belief about productivity. For a given pair (x, y) , the optimal assessment intensity $h(x, y)$ solves

$$\max_{h \in [0, 1]} v_E(a(h, y), x) - \kappa(h).$$

In our model, $y = x$, so the optimal assessment intensity is $h(x, x)$. Under heterogeneous priors, $y = \beta^*$, so the optimal assessment intensity is $h(x, \beta^*)$. While x has a greater influence on h in our case, it is also less distorted by Proposition 3. Overall, we cannot rank assessment intensity, let alone the other equilibrium objects, across the two scenarios.

One distinction remains clear: under heterogeneous priors, S's equilibrium belief about effort choice is accurate, so ability stereotypes do not spill over into the effort-choice domain.

Double misspecification. The case of double misspecification was originally studied in Murooka and Yamamoto (2021, 2023). In our setting, it arises when S mistakenly believes that A shares its first-order belief about the fundamental. Given an initial belief $\hat{\beta}$, S

perceives A's effort choice as $a(h(\hat{\beta}), \hat{\beta})$, and the KL minimization problem becomes

$$\min_{x \in [\underline{\beta}, \bar{\beta}]} \left| \Delta + \underbrace{r(a(h(\hat{\beta}), \hat{\beta}), x)}_{r(a(h(\hat{\beta}), x), x) \text{ in our model}} - r(a(h(\hat{\beta}), \beta^*), \beta^*) \right|.$$

Since $x = \hat{\beta}$ must hold in equilibrium, the set of Berk–NE is the same under double misspecification and in our model. However, the learning foundations of these equilibria differ—a distinction we now turn to.

Learning foundation. Online Appendix C examines the learning foundations of our Berk–NE. To illustrate the key challenges posed by strategic uncertainty, suppose that $r(a, \beta) = \beta a$ as in Applications 1 and 2. At the outset, S holds a prior $p_0(\beta)$ over β that is smoothly supported on $[\underline{\beta}, \bar{\beta}]$. In every period $i \in \mathbb{N}$, E commits to an assessment intensity h_i , A privately exerts effort $a(h_i, \beta^*)$, and an outcome X_i is generated and public realized. Based on the public history $H_n := (h_i, X_i)_{i=1}^n$ of assessments and outcomes, S updates its belief about β to

$$p_n(\beta \mid H_n) = \frac{\exp \left(- \sum_{i=1}^n \frac{(X_i - \hat{\mu} - \beta a(h_i, \beta))^2}{2h_i^{-1}} \right) p_0(\beta)}{\int_{\underline{\beta}}^{\bar{\beta}} \exp \left(- \sum_{i=1}^n \frac{(X_i - \hat{\mu} - \tilde{\beta} a(h_i, \tilde{\beta}))^2}{2h_i^{-1}} \right) p_0(\tilde{\beta}) d\tilde{\beta}},$$

where the exponential term is proportional to the likelihood of the outcomes given β . Crucially, the posterior updating kernel is *not* Gaussian in β , as the likelihood is *not* quadratic in β . This occurs precisely because A's effort choice, as perceived by S, is a random variable of β . Persistent asymmetric information in β yields strategic uncertainty surrounding effort choice, even after conditioning on the public history.

In models without strategic uncertainty, A's effort choice, as perceived by S, is adapted to the public history. For instance, under heterogeneous priors, the period- n effort choice—both real and perceived—equals $a(h_n, \beta^*)$. Under double misspecification, the perceived effort choice in period n solves

$$\max_{a \in [0,1]} \int_{\underline{\beta}}^{\bar{\beta}} h_n [\hat{\mu} + r(a, \beta)] p_{n-1}(\beta \mid H_{n-1}) d\beta - c(a).$$

The solution is again adapted to the public history. Substituting such a term for $a(h, \beta)$ in the posterior formula yields a Gaussian kernel in β . Under specific priors—such as a uniform

or truncated normal prior—the posterior follows a truncated normal distribution.

In Online Appendix C.2, we show that truncated normality yields an $O(1/n)$ approximation error in the stochastic approximation algorithm that describes the update of posterior beliefs. This fast rate is used, for example, by Murooka and Yamamoto (2023) to establish zero-probability convergence to unstable Berk–Nash equilibria. Under strategic uncertainty, the approximation error decays more slowly, but still sufficiently fast to preserve this result; details are in Online Appendix C.3.

5.3 Divergent beliefs between E and M

So far, we have modeled E and M as a single entity sharing the same belief about A. This assumption is intended to capture societal-level discrimination against A, and relaxing it does not fundamentally change the analysis.

Consider a model variant in which each player i ’s prior, $i \in \{E, M\}$, is supported on $\{\hat{\mu}_i\} \times [\underline{\beta}, \bar{\beta}]$, where $\hat{\mu}_i$ may differ across E and M. Under Gaussian signals, M’s belief shapes A’s incentives indirectly through E’s assessment choice. As long as E’s problem retains the same form as earlier, so are the strategies of E and A. A Berk–Nash equilibrium is then characterized by a vector $(\hat{\beta}_E, \hat{\beta}_M)$ of degenerate beliefs, where each $\hat{\beta}_i$ solves player i ’s KL-minimization problem:

$$\min_{x \in [\underline{\beta}, \bar{\beta}]} |\Delta_i + R(h(\hat{\beta}_E), x) - R(h(\hat{\beta}_E), \beta^*)|, \quad \text{where } \Delta_i := \hat{\mu}_i - \mu^*.$$

Clearly, $\hat{\beta}_E$ remains unchanged relative to our model, whereas $\hat{\beta}_M$ passively solves M’s KL minimization problem but plays no active role elsewhere.

6 Related literature

Misspecified Bayesian learning. Misspecified Bayesian learning has long been studied in statistical decision theory (Berk, 1966), with recent economic research emphasizing its interaction with strategic behavior; a seminal reference is Esponda and Pouzo (2016). We make several contributions to this literature.

First, we contribute to the comparative statics of Berk–Nash equilibria. We adopt Lyapunov local asymptotical stability as the stability notion and use the implied out-of-equilibrium dynamics to derive monotone comparative statics for all stable equilibria. Conceptually, this is an application of the correspondence principle (Echenique, 2002, 2006), although our stability notion and learning dynamics are new. Recently, Ghosh (2024) de-

rives robust comparative statics for Berk–Nash equilibria in Markov decision problems, via mechanisms distinctive of ours. [Heidhues et al. \(2018\)](#) study over- and under-confidence in a single-agent decision problem. Theorem 1 establishes a related direction of distortion in a principal-agent environment. The main difference lies in Theorem 2: monotone comparative statics for all stable equilibria.

Second, our setting features strategic uncertainty about the agent’s effort choice. Beyond the multi-agent models discussed in Section 5.2, strategic uncertainty is absent, by construction, from misspecified learning models with a single decision maker ([Heidhues et al., 2018, 2021; Esponda et al., 2021; Fudenberg et al., 2021; Frick et al., 2023](#)). It is also absent from the social-learning models of [Frick et al. \(2020\)](#), [Bohren and Hauser \(2021\)](#), and [Arieli et al. \(2025\)](#), where the main obstacle to information aggregation is misperceived strategies, i.e., misperceived mappings from private signals to public behavior.⁸ In our setting, the agent’s effort strategy is known in equilibrium; still, inaccurate beliefs about productivity generate inaccurate effort inferences.

Third, we provide learning foundations for Lyapunov locally asymptotically stable equilibria, emphasizing the challenges posed by strategic uncertainty. Our approach combines Bayesian statistics with stochastic approximation, with [Benaïm \(1999\)](#) and [Pemantle \(1990\)](#) as the key mathematical references. In economics, stochastic approximation has been used to study misspecified Bayesian learning by [Esponda et al. \(2021\)](#), [Heidhues et al. \(2021\)](#), and [Murooka and Yamamoto \(2021, 2023\)](#); the last three papers examine Gaussian-signal environments without strategic uncertainty.⁹ Alternative, nonparametric approaches exist. For instance, [Fudenberg et al. \(2021\)](#) establish convergence to uniformly strict Berk–Nash equilibria, a refinement that relies on finite actions and outcomes and does not restrict the equilibrium set in our setting with a continuum of actions, outcomes, and states. [Frick et al. \(2023\)](#) also consider Lyapunov asymptotic stability, but assume finite states and a single agent, or closely related structures.

To our best knowledge, we are the first to study misspecified learning in principal–agent relationships. The vast literature on behavioral incentive contracting typically examines how profit-maximizing principals exploit agents’ exogenously given biases, with no learning on either side; see [Köszegi \(2014\)](#) for a survey. In our discrimination applications, it is the principal, not the agent, who holds misspecified beliefs about the agent’s innate ability. [Niu \(2023\)](#) adds to [Heidhues et al. \(2018\)](#) an initial stage in which a principal fixes the

⁸[He and Libgober \(forthcoming\)](#) study strategy misperceptions that arise from misspecifying the correlation between players’ private signals in an evolution-learning model.

⁹Other applications of stochastic approximation to economics include stochastic fictitious play ([Benaïm et al., 2006](#)), evolutionary games ([Sandholm, 2011](#)), and social learning ([Dasaratha and He, 2022; Arieli et al., 2025](#)).

agent’s overconfidence level once and for all. The feedback between the principal’s policy and learning central to our work is therefore absent.

Discrimination. The theoretical literature on discrimination has largely focused on statistical discrimination, exemplified by the Phelpsian (Aigner and Cain, 1977) and Arrowian (Coate and Loury, 1993) traditions. See Fang and Moro (2011) for a survey, and Echenique and Li (2025) for a recent attempt to bridge these traditions. Traditional statistical discrimination models assume that agents hold correct beliefs. Fueled by growing empirical evidence (e.g., Bohren et al. 2025a) and theoretical developments, a growing literature studies the role of misspecified learning in fostering discrimination.¹⁰

Bohren et al. (2019) study a model in which a sequence of evaluators holding an inaccurate prior conducts costly assessments of a minority agent’s quality. The authors establish, both theoretically and experimentally, a reversal of discriminatory beliefs over time that does not arise in traditional models of taste-based or statistical discrimination. However, the minority agent does not exert effort to influence outcomes. The agency relationship and learning about effort productivity—both central to our paper—are absent from their model.

Bohren et al. (2025b) study “systemic discrimination,” i.e., how discrimination in one domain, such as a sector or organizational hierarchy, spills over into others, through belief-based mechanisms distinct from ours.

Heidhues et al. (2025) model how overconfidence about one’s own caliber shapes beliefs about both in-groups and out-groups, emphasizing the interaction between learning and group identity. He et al. (2023) study how estimation of a misspecified statistical model affects predictions, with discrimination as an application. Both examine passive learning, i.e., the decision maker does not take actions to influence the outcome-generating process.

Implicit incentives. The literature on career concerns, or implicit incentive provision through market beliefs is vast. The classical work of Holmström (1999) assumes Gaussian outcomes with mean equal to the sum of ability and effort. Since Holmström (1999), the literature has been extended to allow for general signal distributions (Dewatripont et al., 1999a), multiplicative ability and effort (Dewatripont et al., 1999b), long-lived agents with dynamic incentives (Bonatti and Hörner, 2017; Cisternas, 2018), and the strategic design of signal distributions (Hörner and Lambert, 2021; Saeedi and Shourideh, 2020).

¹⁰A large psychology literature studies stereotypes arising from hardwired, inaccurate beliefs fueled by mental shortcuts and heuristics; see Nelson (2009) for a survey and Bordalo et al. (2016) and Fryer and Jackson (2008) for economic models. However, this literature focuses on non-Bayesian belief updating and abstracts from the feedback loop between beliefs and behavior emphasized in this paper.

The misspecified learning approach is conceptually related to taste-based discrimination (Becker, 2010), insofar as both misspecified priors and tastes are hardwired. But unlike tastes, beliefs are subsequently updated and carry novel implications for equilibria, welfare, and policy design.

Our equilibrium analysis is embedded in single-period, additive-normal environment. Additivity between ability and effort is well suited for deriving our result on ability–effort tradeoff. It also ensures that equilibrium multiplicity arises solely from endogenous misspecified learning, rather than from more traditional channels. The Gaussian assumption yields tractability and exposes the agent to linear incentive strength. Thus our results apply directly to explicit contracting problems in which parties are restricted to using affine contracts in outcomes, as in [Holmstrom and Milgrom \(1987\)](#).

The learning foundation in Online Appendix [C](#) assumes a myopic agent. Thus, the dynamic incentive problems central to the aforementioned papers, and more generally to games with persistent asymmetric information and moral hazard, do not arise. Recent information-design studies examine different economic settings from ours, without misspecified learning.

A Assumptions [1](#) and [2](#)

In this appendix, we provide sufficient conditions for Assumption [1](#) and establish the genericity of Assumption [2](#).

We refer to the following objectives collectively as the *primitives*: A’s effort cost function, E’s assessment cost function, effective effort, and E’s valuation of effort:

$$\begin{aligned} c : \mathbb{R}_+ &\rightarrow \mathbb{R}, & a &\mapsto c(a), & \kappa : [0, 1] &\rightarrow \mathbb{R}, & h &\mapsto \kappa(h), \\ r : \mathbb{R}_+^2 &\rightarrow \mathbb{R}, & (a, \beta) &\mapsto r(a, \beta), & v_E : \mathbb{R}_+^2 &\rightarrow \mathbb{R}, & (a, \beta) &\mapsto v_E(a, \beta). \end{aligned}$$

For r and v_E , we extend their β -argument to $\mathbb{R}_+$.

We begin by imposing standard assumptions on the primitives:

Assumption 3. *All primitives are smooth functions. Moreover,*

- (i) c is strictly increasing and strictly convex, with $c'(0) = 0$ and $\lim_{a \rightarrow +\infty} c'(a) = +\infty$.
- (ii) κ is strictly increasing and strictly convex, with $\kappa'(0) = 0$ and $\lim_{h \rightarrow 1} \kappa'(h) = +\infty$.
- (iii) r is weakly concave in a , has strictly increasing differences in (a, β) , and satisfies:
 $r(0, \beta) \equiv r(a, 0) \equiv 0$; $r_a(0, \beta) > 0$ and $\lim_{a \rightarrow +\infty} \frac{r_a(a, \beta)}{c'(a)} = 0$ for all $\beta > 0$.
- (iv) v_E is strictly increasing in both arguments, weakly concave in a , has strictly increasing differences in (a, β) , and satisfies $v_{E,a}(0, \beta) > 0$ for all $\beta > 0$.

We solve for players’ optimization problems under Assumption [3](#). For each $(h, \beta) \in [0, 1] \times \mathbb{R}_+$, A solves problem [\(1\)](#). We use $a(h, \beta)$ to denote the set of solutions, and refer

to the set-valued map $(h, \beta) \mapsto a(h, \beta)$ as A's effort strategy. The optimal effective effort is $R(h, \beta) \equiv r(a(h, \beta), \beta)$. For each $\beta \geq 0$, E solves problem (2). We denote the set of solutions by $h(\beta)$ and refer to the set-valued map $\beta \mapsto h(\beta)$ as E's assessment strategy.

Proposition 4. *The following hold under Assumption 3:*

- (i) $(h, \beta) \mapsto a(h, \beta)$ is a smooth function that satisfies $a(h, 0) \equiv a(0, \beta) \equiv 0$, is strictly increasing in h for all $\beta > 0$ and in β for all $h \in (0, 1]$.
- (ii) $(h, \beta) \mapsto R(h, \beta)$ satisfies the properties in Part (i).
- (iii) $h(0) = \{0\}$ and $h(\beta) \subseteq (0, 1)$ for all $\beta > 0$.

Proof. Part (i): For every $(h, \beta) \in [0, 1] \times \mathbb{R}_+$, A solves (1):

$$\max_{a \geq 0} hr(a, \beta) - c(a).$$

Since the objective is strictly concave in a , the problem admits a unique solution, which we denote simply by $a(h, \beta)$.

When either $h = 0$ or $\beta = 0$, $hr(a, \beta) = 0$ holds, while $c(a)$ is strictly increasing in a . Therefore, $a(0, \beta) \equiv a(h, 0) \equiv 0$. When $h, \beta > 0$, $hr_a(0, \beta) - c'(0) > 0$ and $\lim_{a \rightarrow +\infty} hr_a(a, \beta) - c'(a) = -\infty$ hold by Assumption 3. Therefore, $a(h, \beta) > 0$ and is determined by the first-order condition:

$$hr_a(a, \beta) = c'(a). \quad (7)$$

The implicit function theorem then implies that $a(h, \beta)$ is smooth in (h, β) . In the interior of the domain, a 's partial derivatives satisfy

$$\frac{\partial a}{\partial h} = \frac{r_a}{c'' - hr_{aa}} \Big|_{a=a(h, \beta)} > 0, \quad \frac{\partial a}{\partial \beta} = \frac{hr_{a\beta}}{c'' - hr_{aa}} \Big|_{a=a(h, \beta)} > 0, \quad (8)$$

where both inequalities follow from Assumption 3 and the fact that $a(h, \beta) > 0$.

Part (ii): $R(h, \beta) = r(a(h, \beta), \beta)$ is smooth in (h, β) . When either $h = 0$ or $\beta = 0$, $a(h, \beta) = 0$ holds by Part (i). The assumption $r(0, \beta) \equiv 0$ then implies that $R(0, \beta) \equiv R(h, 0) \equiv 0$.

In the interior of the domain, R 's partial derivatives satisfy:

$$\frac{\partial R}{\partial h} = \frac{\partial r}{\partial a} \Big|_{a=a(h, \beta)} \frac{\partial a}{\partial h} > 0, \quad \frac{\partial R}{\partial \beta} = \frac{\partial r}{\partial a} \Big|_{a=a(h, \beta)} \frac{\partial a}{\partial \beta} + \frac{\partial r}{\partial \beta} \Big|_{a=a(h, \beta)} > 0, \quad (9)$$

where both inequalities follow again from Assumption 3 and $a(h, \beta) > 0$.

Part (iii): For every $\beta \geq 0$, E solves (2):

$$\max_{h \in [0,1]} V_E(h, \beta) - \kappa(h) \quad \text{where} \quad V_E(h, \beta) = v_E(a(h, \beta), \beta).$$

When $\beta = 0$, $a(h, 0) \equiv 0$ holds by Part (i), so $V_E(h, 0) = v_E(0, 0)$ and is independent of h . Meanwhile, $\kappa(h)$ is strictly increasing in h , so $h = 0$ is the unique solution.

The derivative of the objective w.r.t. h equals

$$v_{E,a}(a(h, \beta), \beta) \frac{\partial a}{\partial h} - \kappa'(h).$$

For a given $\beta > 0$, this derivative is strictly positive at $h = 0$ and diverges to $-\infty$ as $h \uparrow 1$. Therefore, all solutions, collected in the set $h(\beta)$, belong to $(0, 1)$. $\square$

The next assumption—satisfied by Applications 1 and 2 and more—imposes additional structures on the endogenous objects.

Assumption 4. (i) $r(a, \beta) = \beta a$, $c''' \geq 0$, and $c''' \in [0, \frac{(c'')^2}{c'}]$.¹¹

(ii) $v_E(a, \beta) = \lambda_1 r(a, \beta) - \lambda_2 c(a)$ with $\lambda_1 > 0$ and $\lambda_2 \geq 0$.¹²

Proposition 5. Suppose Assumption 3 holds.

(i) If Part (i) of Assumption 4 holds, then $a(h, \beta)$ has weak increasing differences in (h, β) and is weakly concave in h . It follows that:

(a) $R(h, \beta)$ has strict increasing differences in (h, β) .

(b) $V_E(h, \beta)$ is weakly concave in h . Consequently, $\beta \mapsto h(\beta)$ is a smooth function.

(ii) If Part (ii) of Assumption 4 also holds, then $V_E(h, \beta)$ has strict increasing differences in (h, β) , so $h(\beta)$ is strictly increasing in β .

Proof. Part (i): When $r(a, \beta) = \beta a$, solving (1) explicitly yields $a(h, \beta) = (c')^{-1}(h\beta)$, with partial derivatives

$$\frac{\partial a}{\partial h} = \frac{\beta}{c'}, \quad \frac{\partial a}{\partial \beta} = \frac{h}{c'}, \quad \frac{\partial^2 a}{\partial h^2} = -\frac{c'''}{c''} \left(\frac{\partial a}{\partial h} \right)^2, \quad \frac{\partial^2 a}{\partial h \partial \beta} = \frac{1 - \frac{c'''}{c''} \frac{c'}{c''}}{c''}.$$

¹¹The condition on c''' is satisfied by all power functions with exponent at least two and the exponential function. The upper bound is taken to be $+\infty$ when $c' = 0$ (at $a = 0$).

¹²Part (ii) of Assumption 4 ensures that $\frac{\partial^2 v_E}{\partial h \partial \beta} > 0$ through exploiting the alignment between A and E's problems. Alternative sufficient assumptions exist. Indeed, a standard calculation yields

$$\frac{\partial^2 v_E}{\partial h \partial \beta} > 0 \iff \frac{\partial^2 v_E}{\partial a^2} \Big|_{a=a(h, \beta)} > - \left[\frac{c''}{c'} \left(\beta \frac{\partial^2 v_E}{\partial a \partial \beta} + \frac{\partial v_E}{\partial a} \left(1 - \frac{c'''}{c''} \frac{c'}{c''} \right) \right) \right] \Big|_{a=a(h, \beta)}.$$

The latter condition holds when v_E is not overly concave in a .

Under the assumption $c''' \in [0, \frac{(c'')^2}{c'}]$, $\frac{\partial^2 a}{\partial h^2} \leq 0$ and $\frac{\partial^2 a}{\partial h \partial \beta} \geq 0$ hold whenever $h, \beta > 0$.

Part (i-a): The expression of $\frac{\partial R}{\partial h}$ is given in (9). Simplifying it using (7) yields

$$\frac{\partial R}{\partial h} = r_a(a(h, \beta), \beta) \frac{\partial a}{\partial h} = \frac{c'(a(h, \beta))}{h} \frac{\partial a}{\partial h},$$

and taking derivative w.r.t. β yields

$$\frac{\partial^2 R}{\partial h \partial \beta} = \frac{1}{h} \left[c''(a(h, \beta)) \frac{\partial a}{\partial \beta} \frac{\partial a}{\partial h} + c'(a(h, \beta)) \frac{\partial^2 a}{\partial h \partial \beta} \right].$$

When $h, \beta > 0$, the right-hand side is strictly positive. Therefore, $R(h, \beta)$ has strict increasing differences in (h, β) .

Part (i-b): We have

$$\frac{\partial^2 V_E}{\partial h^2} = \frac{\partial^2 v_E}{\partial a^2} \Big|_{a=a(h, \beta)} \left(\frac{\partial a}{\partial h} \right)^2 + \frac{\partial v_E}{\partial a} \Big|_{a=a(h, \beta)} \frac{\partial^2 a}{\partial h^2}.$$

When $h, \beta > 0$ and so $a(h, \beta) > 0$, $\partial^2 V_E / \partial h^2 \leq 0$ holds because $\frac{\partial v_E}{\partial a} \geq 0$ and $\frac{\partial^2 v_E}{\partial a^2} \leq 0$ by assumption, and $\frac{\partial^2 a}{\partial h^2} \leq 0$ holds by Part (i). Subtracting $\kappa(h)$ from $V_E(h, \beta)$ yields E's objective in (2), which is strictly concave in h . All results follow routinely.

Part (ii): We have

$$\frac{\partial V_E}{\partial h} = [\lambda_1 r_a(a(h, \beta), \beta) - \lambda_2 c'(a(h, \beta))] \frac{\partial a}{\partial h} = \left(\frac{\lambda_1}{h} - \lambda_2 \right) c'(a(h, \beta)) \frac{\partial a}{\partial h},$$

where the first equality exploits the functional form of v_E , and the second equality invokes A's first-order condition (7). When $h \geq \frac{\lambda_1}{\lambda_2}$ ($= +\infty$ if $\lambda_2 = 0$), $\frac{\partial V_E}{\partial h}(h, \beta) - \kappa'(h) \leq 0 - \kappa'(h) < 0$. Thus, when solving E's problem (2), it is without loss of generality (w.l.o.g.) to restrict the domain of h to $(0, \min\{1, \lambda_1/\lambda_2\})$. Over this domain,

$$\frac{\partial^2 V_E}{\partial h \partial \beta} = \left(\frac{\lambda_1}{h} - \lambda_2 \right) \left[c''(a(h, \beta)) \frac{\partial a}{\partial \beta} \frac{\partial a}{\partial h} + c'(a(h, \beta)) \frac{\partial^2 a}{\partial h \partial \beta} \right] > 0$$

whenever $\beta > 0$. Therefore, $h(\beta)$ is strictly increasing in β . □

The next proposition establishes the genericity of Assumption 2:

Proposition 6. *Fix any primitives such that $a(\cdot, \cdot)$, $R(\cdot, \cdot)$, and $h(\cdot)$ are smooth functions. Then Assumption 2 holds for generic Δ .*

Proof. Define

$$f(\beta) := R(h(\beta), \beta) - R(h(\beta), \beta^*) \quad \text{for } \beta \geq 0, \quad (10)$$

with derivative w.r.t. β :

$$f'(\beta) = [R_h(h(\beta), \beta) - R_h(h(\beta), \beta^*)] h'(\beta) + R_\beta(h(\beta), \beta).$$

By Sard's lemma, $f(\{\beta : f'(\beta) = 0\})$ has Lebesgue measure zero. That is, there exists a set $C \subset \mathbb{R}$ of measure zero such that for all $\Delta \notin C$, $f'(\beta) \neq 0$ for all $\beta \in f^{-1}(-\Delta)$.

Fix any $\Delta \notin C$, and let $\hat{\beta}$ be a Berk-NE belief given Δ . If $\hat{\beta} \in (\underline{\beta}, \bar{\beta})$, then $\hat{\beta}$ is an SCE belief. It follows that on a small neighborhood of $\hat{\beta}$, Equation (4) in x :

$$R(h(\beta), x) - R(h(\beta), \beta^*) = -\Delta, \quad x \in \mathbb{R}_+$$

has a root $\psi(\beta)$ that satisfies $\psi(\beta) = \tilde{\psi}(\beta)$ and $\psi(\hat{\beta}) = \hat{\beta}$. By the implicit function theorem,

$$\psi'(\beta) = \frac{[R_h(h(\beta), \beta^*) - R_h(h(\beta), \psi(\beta))] h'(\beta)}{R_\beta(h(\beta), \psi(\beta))}$$

on this neighborhood. Evaluating ψ' at $\beta = \hat{\beta}$ then gives

$$\psi'(\hat{\beta}) = \frac{[R_h(h(\hat{\beta}), \beta^*) - R_h(h(\hat{\beta}), \hat{\beta})] h'(\hat{\beta})}{R_\beta(h(\hat{\beta}), \hat{\beta})} \neq 1,$$

where the inequality follows by $f'(\hat{\beta}) \neq 0$.

Now suppose $\hat{\beta} = \bar{\beta}$. By definition, $\tilde{\psi}(\bar{\beta}) = \bar{\beta}$. We claim that $\tilde{\psi}(\beta) \neq \beta$ on a small neighborhood $(\bar{\beta} - \epsilon, \bar{\beta})$, so $\bar{\beta}$ is an isolated fixed point of $\tilde{\psi}$. Suppose, toward a contradiction, that there exists a sequence $\beta_n \uparrow \bar{\beta}$ such that $\tilde{\psi}(\beta_n) = \beta_n$. Since $\beta_n \uparrow \bar{\beta}$, $\tilde{\psi}(\beta_n) = \psi(\beta_n)$ must hold for large n . Thus $\psi(\beta_n) = \beta_n$ holds for large n , and $\psi(\bar{\beta}) = \bar{\beta}$ by continuity. As established above, fixed points of ψ are isolated, a contradiction.

The proof for the case $\hat{\beta} = \underline{\beta}$ is analogous. □

B Omitted proofs from the main text

Proof of Theorem 1. The proof presented in Section 3.4 assumed that for each $\beta \in [\underline{\beta}, \bar{\beta}]$, Equation (4) in x :

$$R(h(\beta), x) - R(h(\beta), \beta^*) = -\Delta, \quad x \in \mathbb{R}_+$$

has a well-defined root, which we denote by $\psi(\beta)$. We now consider situations where this assumption may fail.

Consider Case A: $\Delta < 0$, in which $\tilde{\psi}(\beta)$ —defined as the choice of $x \in [\underline{\beta}, \bar{\beta}]$ that minimizes the gap between the two sides of (4)—must satisfy $\tilde{\psi}(\beta) > \beta^*$. Consider an increase of β from β' to β'' , where $\beta', \beta'' \in [\underline{\beta}, \bar{\beta}]$. Since $h(\cdot)$ is strictly increasing and $R(\cdot, \cdot)$ has strictly increasing differences, the left-hand side of (4) strictly increases for fixed $x > \beta^*$. To minimize the gap between the two sides of (4), $\tilde{\psi}(\beta)$ must decrease, and strictly so whenever $\tilde{\psi}(\beta'') < \bar{\beta}$. The remainder of the proof proceeds exactly as in Section 3.4.

The proof for Case B: $\Delta > 0$, Parts (i) and (ii), follow analogously.

We finally establish Case B Part (iii): for arbitrary $\bar{\beta} > \beta^*$ and $\Delta > 0$, there exists $\check{\beta} \in (0, \beta^*)$ such that whenever $\underline{\beta} \in (0, \check{\beta})$, it is a Berk–NE belief.

To begin, observe that the function f defined in (10) satisfies $f(0) = f(\beta^*) = 0$, and is moreover nonmonotonic and negative on $(0, \beta^*)$. Thus for an arbitrary $\Delta > 0$, there exists $\check{\beta} \in (0, \beta^*)$ such that $f(\beta) > -\Delta$ for all $\beta \in [0, \check{\beta}]$. Pick any $\underline{\beta} \in (0, \check{\beta})$. If $\psi(\underline{\beta})$ is well defined, the proof proceeds exactly as in Section 3.4. If not, then the following must hold: $R(h(\underline{\beta}), 0) - R(h(\underline{\beta}), \beta^*) = 0 - R(h(\underline{\beta}), \beta^*) > -\Delta$. Thus for $\beta = \underline{\beta}$, setting $\tilde{\psi}(\underline{\beta}) = \underline{\beta}$ minimizes the gap between the two sides of (4). $\square$

Proof of Theorem 3. Denote $\{j : \Delta_j > 0\} := \{1, \dots, K\}$ and $\{j : \Delta_j < 0\} := \{K + 1, \dots, J\}$, with $1 < K < J$. Define

$$D := \left(\prod_{j=1}^K [\underline{\beta}, \beta_j^*] \right) \times \left(\prod_{j=K+1}^J [\beta_j^*, \bar{\beta}] \right).$$

Inspecting the KL minimization problem (6) reveals that any solution must lie in D . In what follows, we restrict the domain of β to D .

Parts (i) and (ii): The same argument for Proposition 4 can be replicated to show that for a given $\beta \in D$, $h(\beta)$ —defined as the optimal assessment intensity for β solving (5)—is determined by the first-order condition:

$$\sum_{j=1}^J \alpha_j \frac{\partial V_E}{\partial h}(h, \beta_j) = \kappa'(h).$$

Since V_E has strict increasing differences, we can bound the left-hand side of the first-order condition as

$$\sum_{j=1}^J \alpha_j \frac{\partial V_E}{\partial h}(h, \beta_j) > \sum_{j=K+1}^J \alpha_j \frac{\partial V_E}{\partial h}(h, \beta_j^*).$$

Therefore, $h(\boldsymbol{\beta}) > \check{h}$, where $\check{h}$ is defined as the unique root of

$$\sum_{j=K+1}^J \alpha_j \frac{\partial V_E}{\partial h}(h, \beta_j^*) = \kappa'(h).$$

We emphasize that $\check{h}$ depends only on V_E and $(\alpha_j, \beta_j^*)_{j=K+1}^J$, but not on the vector of mis-specifications $\boldsymbol{\Delta} := (\Delta_j)_{j=1}^J$.

For each $j = 1, \dots, J$ and $\boldsymbol{\beta} \in D$, define $\psi_j(\boldsymbol{\beta})$ as the root of

$$R(h(\boldsymbol{\beta}), x) - R(h(\boldsymbol{\beta}), \beta_j^*) = -\Delta_j, \quad x \in \mathbb{R}_+.$$

When $\|\boldsymbol{\Delta}\|$ is small, where $\|\cdot\|$ denotes the Euclidean norm, the existence and uniqueness of the root are clear. For each $j = K+1, \dots, J$ such that $\Delta_j < 0$, we have

$$-\Delta_j = R(h(\boldsymbol{\beta}), \psi_j(\boldsymbol{\beta})) - R(h(\boldsymbol{\beta}), \beta_j^*) > R(\check{h}, \psi_j(\boldsymbol{\beta})) - R(\check{h}, \beta_j^*) > 0,$$

where the inequalities hold because $R(h, \beta)$ has strict increasing differences, $h(\boldsymbol{\beta}) > \check{h}$, and $\psi_j(\boldsymbol{\beta}) > \beta_j^*$. It follows that $\psi_j(\boldsymbol{\beta}) - \beta_j^* = O(|\Delta_j|)$. The argument for $j = 1, \dots, K$, for whom $\Delta_j > 0$, is analogous. Denoting $\boldsymbol{\psi}(\boldsymbol{\beta}) := (\psi_j(\boldsymbol{\beta}))_{j=1}^J$, we have $\boldsymbol{\psi}(\boldsymbol{\beta}) \in D$, so the function $\boldsymbol{\beta} \mapsto \boldsymbol{\psi}(\boldsymbol{\beta})$ maps D to itself.

By the implicit function theorem,

$$\nabla_{\boldsymbol{\beta}} \boldsymbol{\psi} = \mathbf{g}(\boldsymbol{\beta}; \boldsymbol{\Delta}) (\nabla_{\boldsymbol{\beta}} h)^\top \quad \text{where}$$

$$\mathbf{g}(\boldsymbol{\beta}; \boldsymbol{\Delta}) := (g_j(\boldsymbol{\beta}; \Delta_j))_{j=1}^J, \quad g_j(\boldsymbol{\beta}; \Delta_j) := \frac{\frac{\partial R}{\partial h}(h(\boldsymbol{\beta}), \beta_j^*) - \frac{\partial R}{\partial h}(h(\boldsymbol{\beta}), \psi_j(\boldsymbol{\beta}))}{\frac{\partial R}{\partial \beta}(h(\boldsymbol{\beta}), \psi_j(\boldsymbol{\beta}))}.$$

We parameterize $\mathbf{g}$ by $\boldsymbol{\Delta}$ because $\boldsymbol{\psi}(\boldsymbol{\beta})$ depends on it but $h(\boldsymbol{\Delta})$ doesn't. The above derivation implies that $g_j(\boldsymbol{\beta}; \Delta_j) = O(|\Delta_j|)$, with $\text{sgn}(g_j(\boldsymbol{\beta}; \Delta_j)) = \text{sgn}(\Delta_j)$.

Since the Jacobian $\nabla_{\boldsymbol{\beta}} \boldsymbol{\psi}$ has rank one, its spectral norm equals the product of the Euclidean norms of its components:

$$\|\nabla_{\boldsymbol{\beta}} \boldsymbol{\psi}\|_2 = \|\mathbf{g}(\boldsymbol{\beta}; \boldsymbol{\Delta})\| \|\nabla_{\boldsymbol{\beta}} h\|.$$

Moreover, since $\|\mathbf{g}(\boldsymbol{\beta}; \boldsymbol{\Delta})\| = O(\|\boldsymbol{\Delta}\|)$, and $\sup_{\boldsymbol{\beta} \in D} \|\nabla_{\boldsymbol{\beta}} h\|$ is finite and independent of $\boldsymbol{\Delta}$, the right-hand side vanishes as $\|\boldsymbol{\Delta}\| \rightarrow 0$. Thus for sufficiently small $\|\boldsymbol{\Delta}\|$, the following are true. First, $\boldsymbol{\psi}$ is a contraction on D ; its unique fixed point, denoted by $\hat{\boldsymbol{\beta}}$, corresponds to the Berk-NE (also SCE) of the game. Second, $\boldsymbol{\Psi}(\boldsymbol{\beta}) := \boldsymbol{\psi}(\boldsymbol{\beta}) - \boldsymbol{\beta}$ has Jacobian $\nabla_{\boldsymbol{\beta}} \boldsymbol{\Psi} = -Id_J + \mathbf{g}(\boldsymbol{\beta}; \boldsymbol{\Delta}) (\nabla_{\boldsymbol{\beta}} h)^\top$, where Id_J denotes the $J \times J$ identity matrix. Since this Jacobian is

a rank-one perturbation of the identity matrix, solving its spectrum yields $J - 1$ eigenvalues equal to -1 , and one eigenvalue equal to $-1 + (\nabla_{\beta} h)^{\top} \mathbf{g}(\beta; \Delta)$. Since all eigenvalues are real and negative, the Berk–NE is locally asymptotically stable in the Lyapunov sense.

Parts (iii) and (iv): We model shocks to the environment as perturbations of a parameter $\rho \in \mathbb{R}$ (interpretations given soon). Parameterize all objects: $\mathbf{g}(\beta; \rho)$, $\psi(\beta; \rho)$, $\Psi(\beta; \rho)$, and $h(\beta; \rho)$ by ρ , and omit—for notational ease—the dependence of all except h on Δ .

We have $\Psi(\hat{\beta}(\rho), \rho) = \mathbf{0}$ and, by the implicit function theorem,

$$\frac{\partial \hat{\beta}}{\partial \rho} = - (\nabla_{\beta} \Psi(\beta; \rho))^{-1} \frac{\partial \Psi}{\partial \rho}(\beta; \rho) \Big|_{\beta=\hat{\beta}(\rho)}.$$

Since $\nabla_{\beta} \Psi$, solved above, is a rank-one perturbation of the identity matrix, applying the Sherman-Morrison formula (stated below) gives

$$(\nabla_{\beta} \Psi(\beta; \rho))^{-1} = -Id_J - \frac{\mathbf{g}(\beta; \rho) (\nabla_{\beta} h(\beta; \rho))^{\top}}{1 - (\nabla_{\beta} h(\beta; \rho))^{\top} \mathbf{g}(\beta; \rho)}$$

and consequently

$$\frac{\partial \hat{\beta}}{\partial \rho} = \left(Id_J + \frac{\mathbf{g}(\beta; \rho) (\nabla_{\beta} h(\beta; \rho))^{\top}}{1 - (\nabla_{\beta} h(\beta; \rho))^{\top} \mathbf{g}(\beta; \rho)} \right) \frac{\partial \Psi}{\partial \rho}(\beta; \rho) \Big|_{\beta=\hat{\beta}(\rho)}.$$

Fact 1 (Sherman and Morrison, 1950). *Let A be a $J \times J$ invertible matrix, and let u and v be J -vectors. Then $A + uv^{\top}$ is invertible if and only if $1 + v^{\top} A^{-1} u \neq 0$, in which case*

$$(A + uv^{\top})^{-1} = A^{-1} - \frac{A^{-1} uv^{\top} A^{-1}}{1 + v^{\top} A^{-1} u}.$$

Consider first a perturbation of $\rho = \Delta_1$, where the subscript “1” denotes coordinate-1 of a vector. We have $\frac{\partial \Psi}{\partial \Delta_1} = \frac{\partial \psi}{\partial \Delta_1}$, where $\frac{\partial \psi_1}{\partial \Delta_1} < 0$ by the same argument for Theorem 1(i), while $\frac{\partial \psi_j}{\partial \Delta_1} = 0$ for $j \neq 1$ by the definition of ψ_j . Substituting this observation into $\frac{\partial \hat{\beta}}{\partial \Delta_1}$ yields $\frac{\partial \hat{\beta}_1}{\partial \Delta_1} < 0$ and $\frac{\partial \hat{\beta}_j}{\partial \Delta_1} \approx 0$ for $j \neq 1$ when $\Delta \approx \mathbf{0}$.

Consider next a perturbation of $\rho = \zeta$. Straightforward algebra shows that

$$\frac{\partial \Psi}{\partial \zeta}(\beta; \zeta) = \frac{\partial h}{\partial \zeta}(\beta; \zeta) \mathbf{g}(\beta; \zeta)$$

and consequently

$$\begin{aligned}\frac{\partial \hat{\beta}}{\partial \zeta} &= \frac{\partial h}{\partial \zeta}(\beta; \zeta) \left(Id_J + \frac{\mathbf{g}(\beta; \zeta)(\nabla_{\beta} h(\beta; \zeta))^{\top}}{1 - (\nabla_{\beta} h(\beta; \zeta))^{\top} \mathbf{g}(\beta; \zeta)} \right) \mathbf{g}(\beta; \zeta) \Big|_{\beta=\hat{\beta}(\zeta)} \\ &= \frac{\partial h}{\partial \zeta}(\beta; \zeta) \frac{1}{1 - (\nabla_{\beta} h(\beta; \zeta))^{\top} \mathbf{g}(\beta; \zeta)} \mathbf{g}(\beta; \zeta) \Big|_{\beta=\hat{\beta}(\zeta)}.\end{aligned}$$

On the right-hand side, $\frac{\partial h}{\partial \zeta} > 0$ because the objective in (5) has strictly increasing differences in (h, ζ) . For $\Delta \approx \mathbf{0}$, the fraction is close to one, and $\text{sgn}(g_j(\cdot)) = \text{sgn}(\Delta_j)$, both as noted earlier. It follows that $\text{sgn}\left(\frac{\partial \hat{\beta}_j}{\partial \zeta}\right) = \text{sgn}(\Delta_j)$. $\square$

Proof of Proposition 1. We only prove Part (iv) of the proposition. Fix an SCE where group i 's productivity is perceived by S as $\hat{\beta}$. Then group i is assessed at intensity $h(\hat{\beta})$, and their effort choice is perceived by S as $\hat{a} = a(h(\hat{\beta}), \hat{\beta})$. Conditional on observing X , M rewards the group by

$$\mathbb{E}_{\delta_{(\hat{\mu}, \hat{\beta}, \hat{a})}}[\eta + v_M(a, \beta) \mid X] = \hat{\mu} + h(\hat{\beta}) [X - \hat{\mu} - R(h(\hat{\beta}), \hat{\beta})] + v_M(\hat{a}, \hat{\beta}).$$

The true mean of X equals $\mu^* + R(h(\hat{\beta}), \beta^*)$, which equals $\hat{\mu} + R(h(\hat{\beta}), \hat{\beta})$ by the definition of SCE. Therefore, $X - \hat{\mu} - R(h(\hat{\beta}), \hat{\beta})$ has true mean zero. $\square$

Proof of Proposition 3. In Section 5, we used δ_x to denote E's first-order belief about A's productivity, and δ_{δ_y} to denote E's second-order belief about A's first-order belief about their own productivity. Setting $y = x$ and $y = \beta^*$ yields our model and the model of heterogeneous priors, respectively. For a given pair $(x, y) \in \mathbb{R}_+^2$, E solves

$$\max_{h \in [0, 1]} v_E(a(h, y), x) - \kappa(h).$$

Under Assumption 3, the solution—denoted by $h(x, y)$ —is unique, lies in $(0, 1)$ if $x, y > 0$, and is strictly increasing in x for fixed $y > 0$ (algebra omitted).

Define

$$f : (x, y) \mapsto r(a(h(x, y), y), x) - r(a(h(x, y), \beta^*), \beta^*), \quad (x, y) \in \mathbb{R}_+^2.$$

Observe that: (i) $f(\beta^*, \beta^*) = 0$; (ii) $f(x, \beta^*), f(x, x) < 0$ on $(0, \beta^*)$; (iii) $f(x, \beta^*), f(x, x) > 0$ on $(\beta^*, +\infty)$. Thus, both functions $x \mapsto f(x, \beta^*)$ and $x \mapsto f(x, x)$ are increasing in x near $x = \beta^*$. For small $|\Delta|$, the roots $f(x, x) = -\Delta$ and $f(x, \beta^*) = -\Delta$ closest to β^* correspond to the least distorted SCE beliefs in our model and in the heterogeneous-prior

model, respectively. To compare these roots, we differentiate $f(x, y)$ with respect to y :

$$\begin{aligned} \frac{\partial f}{\partial y} = & \left[r_a(a(h(x, y), y), x) \frac{\partial a}{\partial h}(h(x, y), y) - r_a(a(h(x, y), \beta^*), \beta^*) \frac{\partial a}{\partial h}(h(x, y), \beta^*) \right] \frac{\partial h}{\partial y} \\ & + r_a(a(h(x, y), y), x) \frac{\partial a}{\partial \beta}(h(x, y), y). \end{aligned}$$

When $(x, y) \approx (\beta^*, \beta^*)$, the first line on the right-hand side is negligible, whereas the second line is strictly positive. Therefore, $f(x, x)$ increases faster than $f(x, \beta^*)$ in a small neighborhood of $x = \beta^*$, from which the desired results follow. $\square$

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Online Appendix for
“Implicit Incentive Provision with Misspecified
Learning”
by Federico Echenique and Anqi Li

C Learning foundation

This appendix develops the misspecified Bayesian learning foundation of the Berk–Nash equilibria studied in the main text. Throughout, we refer to the model in the main text as the *equilibrium model*, and to the model in this appendix as the *learning model*.

Learning model. Time is discrete and infinite, indexed by $n \in \mathbb{N}_0 = \{0\} \cup \mathbb{N}$, and all players are Bayesian. A fundamental $\omega = (\mu, \beta)$ is a vector of mean innate ability μ and effort productivity β . At time 0, the true fundamental $\omega^* = (\mu^*, \omega^*)$ is realized once and for all and is privately observed by A; this information structure is common knowledge. S—comprised of E and M—is uncertain about the fundamental and holds a prior about it. This prior is degenerate at $\hat{\mu} \neq \mu^*$ along the μ -coordinate. Along the β -coordinate, the prior admits a smooth, full-support density p_0 on $[\underline{\beta}, \bar{\beta}] \subset \mathbb{R}_{++}$, with $\beta^* \in (\underline{\beta}, \bar{\beta})$.

In every period $n \in \mathbb{N}$, the following events unfold:

1. E publicly commits to an assessment intensity $h_n \in [0, 1]$.
2. A privately chooses an effort level $a_n \geq 0$.
3. The period- n ability $\eta_n \sim \mathcal{N}(\mu^*, 1)$ and noise $\varepsilon_n \sim \mathcal{N}(0, h_n^{-1} - 1)$ are realized, where $\eta_n - \mu^*$ and ε_n are independent of each other, over time, and from ω^* . Together, they determine the period- n outcome as

$$X_n = \eta_n + r(a, \beta^*) + \varepsilon_n.$$

Only the realization of X_n (but not of η_n or ε_n) becomes public.

4. M forms an expectation of A's period- n value $\eta_n + v_M(a_n, \beta)$ and rewards A accordingly.
5. S updates its belief about β .

All players are myopic and maximize expected payoffs period by period. This assumption is standard in the misspecified-learning literature and relevant in our applications, where society accumulates institutional knowledge about the fundamental, while individual members—students, teachers, workers, and employers—are short-lived.

Let $p_n(\cdot \mid H_n)$ denote the posterior density of β in Stage 5 of period n , formed based on the public history $H_n := (X_i, h_i)_{i=1}^n$. For notation simplicity, we omit the dependence of p_n on H_n whenever the context is clear.

Consider the stage game in period n . Fixing h_n , let $\hat{a}(h_n, \beta)$ denote an arbitrary societal perception about A's effort choice as a random variable of $\beta \sim p_{n-1}(\cdot \mid H_{n-1})$. Then, upon

observing X_n , M's conditional expectation of A's period- n value, for a fixed β , equals

$$\mathbb{E}[\eta_n + v_M(\hat{a}(h_n, \beta), \beta) \mid h_n, X_n] = \hat{\mu} + h_n [X_n - \hat{\mu} - r(\hat{a}(h_n, \beta), \beta)] + v_M(\hat{a}(h_n, \beta), \beta).$$

Taking expectation over $\beta \sim p_{n-1}(\cdot \mid H_{n-1})$ while treating h_n and X_n as observables gives M's reward to A. Crucially, this reward takes the form

$$h_n X_n + \text{terms unaffected by A's actual effort choice.}$$

Because A's actual effort choice influences the mean of X_n , we can follow the derivation in Section 2.1 to obtain $a(h_n, \beta^*)$ as A's optimal effort choice.

In Stage 1, E anticipates that A chooses according to $a(h_n, \cdot)$ given h_n . Setting $\hat{a}(\cdot, \cdot) = a(\cdot, \cdot)$ and doing backward induction yields E's problem in period n as

$$\max_{h \in [\underline{\beta}, \bar{\beta}]} \int_{\underline{\beta}}^{\bar{\beta}} V_E(h, \beta) p_{n-1}(\beta \mid H_{n-1}) d\beta - \kappa(h). \quad (11)$$

Under Assumption 3, (11) admits a unique solution, which we denote by h_n and note its adaptability to H_{n-1} . Since p_{n-1} first-order stochastically dominates $\delta_{\underline{\beta}}$ and is in turn dominated by $\delta_{\bar{\beta}}$, $h_n \in [h(\underline{\beta}), h(\bar{\beta})]$ always holds. For notation simplicity, we write $[\underline{h}, \bar{h}]$ for $[h(\underline{\beta}), h(\bar{\beta})]$ and note that $[\underline{h}, \bar{h}] \subset (0, 1)$.

Consider next the updating of beliefs. In Stage 5 of period n , the public history updates to $H_n = (h_i, X_i)_{i=1}^n$. Applying Bayes' rule yields

$$p_n(\beta \mid H_n) = \frac{\exp\left(-\sum_{i=1}^n \frac{(X_i - \hat{\mu} - R(h_i, \beta))^2}{2h_i^{-1}}\right) p_0(\beta)}{\int_{\underline{\beta}}^{\bar{\beta}} \exp\left(-\sum_{i=1}^n \frac{(X_i - \hat{\mu} - R(h_i, \tilde{\beta}))^2}{2h_i^{-1}}\right) p_0(\tilde{\beta}) d\tilde{\beta}} \quad (12)$$

as the posterior density of β given H_n . The exponents in the numerator and the denominator represent the log-likelihoods of the outcomes given β . In the presence of strategic uncertainty—as discussed in Section 5.2—these terms are typically not quadratic in β , so the updating kernel is *not* Gaussian in β .

We make two additional assumptions about the primitives beyond Assumptions 1 and 2. The first is a mild regularity condition ensuring that $\psi(\beta)$ —the root of $R(h(\beta), x) - R(h(\beta), \beta^*) = -\Delta$ —is well-defined in $\mathbb{R}_+$ for all $\beta \in [\underline{\beta}, \bar{\beta}]$:

Assumption 5. $-(\lim_{x \rightarrow +\infty} R(h(\underline{\beta}), x) - R(h(\underline{\beta}), \beta^*)) < \Delta < R(h(\underline{\beta}), \beta^*)$

The derivation of why Assumption 5 gives rise to well-defined $\psi(\beta)$ is standard and omitted.¹³

The second assumption is more significant, and its role will soon be discussed.

Assumption 6. $R(h, \beta) = g_1(\beta)g_2(h) + g_3(h)$, where both $g_1 : \mathbb{R}_+ \rightarrow \mathbb{R}$ and $g_2 : [0, 1] \rightarrow \mathbb{R}$ are strictly increasing and strictly positive on the interiors of their respective domains.

In Applications 1 and 2, if we maintain $r(a, \beta) = \beta a$ but relax quadratic effort cost to $c(a) = a^\gamma/\gamma$ for some $\gamma > 1$, we would obtain $a(h, \beta) = (h\beta)^{\frac{1}{\gamma-1}}$ and $R(h, \beta) = h^{\frac{1}{\gamma-1}}\beta^{\frac{\gamma}{\gamma-1}}$. Assumption 6 is satisfied, with $g_1(\beta) = \beta^{\frac{\gamma}{\gamma-1}}$, $g_2(h) = h^{\frac{1}{\gamma-1}}$, and $g_3(h) = 0$.

Convergence theorem. We state our main convergence theorem:

Theorem 4. *Let p_0 be as described above, and suppose that Assumptions 1, 2, 5, and 6 hold. As $n \rightarrow \infty$:*

- (i) *S's posterior beliefs about β almost surely converge to the set of the locally asymptotically stable $\hat{\beta}$'s in the equilibrium model.*
- (ii) *For every such $\hat{\beta}$, convergence occurs with strictly positive probability, and with probability arbitrarily close to one given a sufficiently concentrated prior around $\hat{\beta}$.*

The remainder of this appendix is devoted to proving Theorem 4.

Proof overview. In Section 5.2 of the main text, we already discussed a major challenge faced in our analysis: strategic uncertainty, manifested as a lack of adaptability of perceived effort choice to S's public history. Existing works on misspecified Bayesian learning with Gaussian signals reviewed in Section 6—e.g., Heidhues et al. (2021) and Murooka and Yamamoto (2021, 2023)—abstract away from strategic uncertainty and assumes a quadratic log-likelihood in β . Under specific priors such as uniform or (truncated) normal priors, posteriors follow (truncated) normal distributions, and their asymptotic behavior can be analyzed using stochastic approximation.

Assumption 6 is introduced to address this challenge. By stipulating that the optimal effective effort is affine in $\check{\beta} := g_1(\beta)$, it enables a change of variable from β to $\check{\beta}$. Since g_1 is strictly increasing, $g_1^{-1}(\check{\beta})$ is unique for all $\check{\beta} \in g_1([\underline{\beta}, \bar{\beta}])$. When players' problems are expressed in terms of $\check{\beta}$, all analysis in the equilibrium model carries over directly.

¹³In the context of Applications 1 and 2, it can be verified that the first inequality holds automatically, whereas the second inequality is compatible with the condition in Theorem 1 Part (iii), which ensures that the smallest Berk-NE is a boundary equilibrium rather than an SCE when $\Delta > 0$. Indeed, jointly solving both conditions yields $(\beta^*)^2 - (\underline{\beta})^2 < \Delta c \frac{\lambda_2(\underline{\beta})^2 + \kappa c}{\lambda_1(\underline{\beta})^2} < (\beta^*)^2$. Holding all else constant, the whole chain of inequalities can be satisfied by an appropriate choice of Δ .

In the learning model presented in this appendix, this transformation allows us to express the log-likelihood of the outcome as a quadratic function of $\check{\beta}$ (details to follow). The caveat, however, is that the prior on $\check{\beta}$ can be arbitrary, even if the original prior on β is uniform or (truncated) normal. Still, the objective is clear: to reduce the challenges posed by strategic uncertainty to a more primitive question, namely, how does a general prior affect posteriors?

In strategic learning games, the prior influences posteriors directly through Bayesian updating and indirectly through players' strategic behavior. Using Bayesian statistics tools such as Laplace's approximation, we show that the overall impact decays sufficiently fast in time, so that existing stochastic approximation results continue to apply after relevant adaptations.

The remainder of this appendix is organized as follows:

- Appendix C.1 introduces the relevant background on Laplace's approximation.
- Appendix C.2 studies benchmark priors for which posteriors are exactly truncated normal. We establish an analog of Theorem 4 using stochastic approximation, with Benaïm (1999) (B99), Pemantle (1990) (P90), and Murooka and Yamamoto (2023) (MY23) as key mathematical references.
- Appendix C.3 extends the analysis in Appendix C.2 to general priors.
- Appendices C.4 and C.5 collect mathematical proofs.
- Appendix C.6 discusses an alternative to Assumption 6.

For notation simplicity, we relabel $\check{\beta}$ as β and the prior of $\check{\beta}$ as p_0 , with support $[\underline{\beta}, \overline{\beta}]$. We adopt standard matrix notation: $\|\cdot\|$ denotes the Euclidean norm, v_i denotes the i^{th} coordinate of a vector v , and Id denotes the identity matrix.

C.1 Laplace's approximation

Let $\bar{p}_0$ be a smooth density function supported on $\mathbb{R}$ that satisfies

$$p_0(\beta) = \frac{\bar{p}_0(\beta) \mathbf{1}_{\beta \in [\underline{\beta}, \overline{\beta}]}}{\int \bar{p}_0(\tilde{\beta}) \mathbf{1}_{\tilde{\beta} \in [\underline{\beta}, \overline{\beta}]} d\tilde{\beta}} \quad \text{for } \beta \in [\underline{\beta}, \overline{\beta}].$$

Under Assumption 6, for $n \in \mathbb{N}$:

$$\log \bar{p}_n(\beta \mid H_n) = \sum_{i=1}^n -\frac{(X_i - \hat{\mu} - \beta g_2(h_i) - g_3(h_i))^2}{2h_i^{-1}} + \log \bar{p}_0(\beta) + \text{terms independent of } \beta,$$

with the following derivatives w.r.t. β :

$$\begin{aligned}\frac{d}{d\beta} \log \bar{p}_n(\beta \mid H_n) &= \sum_{i=1}^n \frac{X_i - \hat{\mu} - \beta g_2(h_i) - g_3(h_i)}{h_i^{-1}} g_2(h_i) + \frac{d \log \bar{p}_0}{d\beta}, \\ \frac{d^2}{d\beta^2} \log \bar{p}_n(\beta \mid H_n) &= - \sum_{i=1}^n g_2^2(h_i) h_i + \frac{d^2 \log \bar{p}_0}{d\beta^2}, \text{ and} \\ \frac{d^3}{d\beta^3} \log \bar{p}_n(\beta \mid H_n) &= \frac{d^3 \log \bar{p}_0}{d\beta^3}.\end{aligned}$$

The third derivative is independent of n , indicating a negligible impact of derivatives beyond the second order when n is large.

The term $g_2^2(h_n)h_n := I(h_n)$ —or simply I_n —represents the *Fisher information* of the period- n outcome X_n as a signal of β . Since $h_n \geq \underline{h} > 0$, the assumption on g_2 implies that $I_n \geq I(\underline{h}) > 0$. The economic intuition is that, since assessment and the induced effort are always positive, the outcome always reveals new information about β . In what follows, we write $[\underline{I}, \bar{I}]$ for $[I(\underline{h}), I(\bar{h})]$ and note that $\underline{I} > 0$.

Let m_n denote the mode of $\bar{p}_n$. For large n , m_n is determined by the first-order condition:

$$\frac{d}{d\beta} \log \bar{p}_n(\beta \mid H_n) = 0$$

and can be solved explicitly as

$$m_n = \frac{\sum_{i=1}^n (X_i - \hat{\mu} - g_3(h_i)) h_i g_2(h_i) + \frac{d \log \bar{p}_0}{d\beta} \Big|_{\beta=m_n}}{\sum_{i=1}^n I_i}. \quad (13)$$

Evaluating the second derivative of $\log \bar{p}_n(\beta \mid H_n)$ at m_n then gives

$$\frac{d^2}{d\beta^2} \log \bar{p}_n(\beta \mid H_n) \Big|_{\beta=m_n} = - \sum_{i=1}^n I_i + \frac{d^2 \log \bar{p}_0}{d\beta^2} \Big|_{\beta=m_n} := -\frac{1}{v_n}. \quad (14)$$

Based on (13) and (14), we can expand $\log \bar{p}_n(\beta \mid H_n)$ around m_n as follows:

$$\begin{aligned} \log \bar{p}_n(\beta \mid H_n) &= \log \bar{p}_n(m_n \mid H_n) + 0 (\beta - m_n) \\ &\quad + \frac{1}{2} \frac{d^2}{d\beta^2} \log \bar{p}_n(\beta \mid H_n) \Big|_{\beta=m_n} (\beta - m_n)^2 \\ &\quad + \frac{1}{6} \frac{d^3}{d\beta^3} \log \bar{p}_0(\beta) \Big|_{\beta=z(\beta, m_n)} (\beta - m_n)^3 \\ &= -\frac{(\beta - m_n)^2}{2v_n} + K(\beta, m_n)(\beta - m_n)^3 + \text{terms independent of } \beta, \end{aligned}$$

where

$$K(\beta, m_n) := \frac{1}{6} \frac{d^3}{d\beta^3} \log \bar{p}_0(\beta) \Big|_{\beta=z(\beta, m_n)}. \quad (15)$$

Exponentiating both sides yields

$$\bar{p}_n(\beta \mid H_n) \propto \exp \left(-\frac{(\beta - m_n)^2}{2v_n} \right) \exp (K(\beta, m_n)(\beta - m_n)^3),$$

where $\propto$ indicates “proportional to.”

On $[\underline{\beta}, \bar{\beta}]$, $p_0(\beta) \propto \bar{p}_0(\beta)$ holds, so

$$p_n(\beta \mid H_n) \propto \exp \left(-\frac{(\beta - m_n)^2}{2v_n} \right) \exp (K(\beta, m_n)(\beta - m_n)^3).$$

On the right-hand side, the first exponential term corresponds to a normal density with mean m_n and variance v_n . The second is an approximation error that arises from the prior. In Appendix C.5, we show that the process $\{m_n\}$ is almost surely (a.s.) bounded. It follows that for large n , $v_n = \Theta(1/n)$ and $K(\beta, m_n) = O(1)$ a.s., so the second exponential term is negligible relative to the first. Using this fact, we approximate the asymptotic posterior distribution of β by a truncated normal distribution $\tilde{\mathcal{N}}(m_n, v_n)$. The corresponding untruncated normal distribution is $\mathcal{N}(m_n, v_n)$.

C.2 Stochastic approximation: Specific priors

This appendix establishes Theorem 4 under a uniform prior over $[\underline{\beta}, \bar{\beta}]$. The analysis extends directly to truncated normal priors.

The state and its evolution. Under the uniform prior, all derivatives of $\bar{p}_0$ w.r.t. β equal zero. Setting $d^k \bar{p}_0 / d\beta^k = 0$ for $k = 1, 2, 3$ in (13)-(15), the period- n posterior, $n \in \mathbb{N}$, follows exactly a truncated normal distribution $\tilde{\mathcal{N}}(m_n, v_n)$, where m_n and v_n obtain from

omitting terms that involve $\bar{p}_0$ from (13) and (14). For the corresponding untruncated normal distribution $\mathcal{N}(m_n, v_n)$, define

$$\xi_n := \frac{1}{nv_n}$$

as the inverse of the time-scaled variance. The vector $\theta_n := (m_n, \xi_n)$ is the period- n state, which, together with the calendar time n , fully parameterize the period- n posterior distribution. The posterior, in turn, fully determines the next-period assessment h_{n+1} and Fisher information I_{n+1} index. Later, we will demonstrate that $\theta_n \in \Xi := \mathbb{R} \times [\underline{I}, \bar{I}]$, and refer to Ξ as the *state space*.

We describe the evolution of the state through a system of stochastic difference equations. To begin, rewrite (14) as

$$\xi_{n+1} - \xi_n = \frac{1}{n+1} (I_{n+1} - \xi_n). \quad (16)$$

Since $I_n \in [\underline{I}, \bar{I}]$ always holds, the process $\{\xi_n\}$ is uniformly bounded in $[\underline{I}, \bar{I}]$.

Next, recall that $X_n - \hat{\mu} = R(h_n, \beta(h_n)) + \epsilon_n / \sqrt{h_n}$, where $\beta(h_n)$ denotes the unconstrained best-fit parameter of the outcome distribution induced by h_n ; Assumption 5 ensures that it is always well-defined. Projecting $\beta(h_n)$ onto $[\underline{\beta}, \bar{\beta}]$ yields the constrained best-fit parameter $\tilde{\beta}(h_n)$, a central concept in the equilibrium model. Meanwhile, $\epsilon_n := \sqrt{h_n}(\eta_n - \hat{\mu} + \varepsilon_n)$, which is standard normal and i.i.d. over time. Substituting these findings into (13), straightforward algebra yields

$$m_{n+1} - m_n = \frac{1}{n+1} \left(\frac{I_{n+1} (\beta(h_{n+1}) - m_n) + \sqrt{I_{n+1}} \epsilon_{n+1}}{\frac{n}{n+1} \xi_n + \frac{1}{n+1} I_{n+1}} \right). \quad (17)$$

In (16) and (17), write $h_{n+1}(\theta_n)$ for h_{n+1} and $(I \circ h_{n+1})(\theta_n)$ for I_{n+1} to make their dependence on θ_n clear. This gives, in vector form, a stochastic difference equation:

$$\theta_{n+1} - \theta_n = \frac{1}{n+1} \left(F_{n+1}(\theta_n) + b_{n+1}(\theta_n) \epsilon_{n+1} \right), \quad (18)$$

where

$$F_{n+1}(\theta) := \begin{bmatrix} \frac{(I \circ h_{n+1})(\theta) [(\beta \circ h_{n+1})(\theta) - m]}{\frac{n}{n+1} \xi + \frac{1}{n+1} (I \circ h_{n+1})(\theta)} \\ (I \circ h_{n+1})(\theta) - \xi \end{bmatrix}, \quad b_{n+1}(\theta) := \begin{bmatrix} \frac{\sqrt{(I \circ h_{n+1})(\theta)}}{\frac{n}{n+1} \xi + \frac{1}{n+1} (I \circ h_{n+1})(\theta)} \\ 0 \end{bmatrix}.$$

Limiting ODE. When n is large, $v_n = \Theta(1/n)$ and is small. As a result, we may approximate the truncated normal distribution $\tilde{\mathcal{N}}(m_n, v_n)$ by the degenerate distribution $\delta_{\tilde{m}_n}$, where $\tilde{m}_n$ denotes the projection of m_n onto $[\underline{\beta}, \bar{\beta}]$. The optimal assessment given $\delta_{\tilde{m}_n}$ is

$h(\tilde{m}_n)$, where $h(\cdot)$ is E's optimal assessment strategy in the equilibrium model. In (18), replace $h_{n+1}(\theta_n)$ with $h(\tilde{m}_n)$, drop the noise term ϵ_n , and take $n \rightarrow \infty$. This yields an ODE system:

$$\dot{\theta} = F(\theta) := \begin{bmatrix} \frac{(I \circ h)(\tilde{m}) (\psi(\tilde{m}) - m)}{\xi} \\ (I \circ h)(\tilde{m}) - \xi \end{bmatrix}, \quad (19)$$

where $\psi(\tilde{m}) = (\beta \circ h)(\tilde{m})$ holds because both sides represent the unconstrained best-fit productivity parameter of the outcome distribution induced by $h(\tilde{m})$.

There is a one-to-one correspondence between the steady states of the limiting ODE and the Berk–Nash equilibria of the equilibrium model. This is because a steady state $(\hat{m}, \hat{\xi})$ of the ODE— $\hat{\xi}$ fully determined by $\hat{m}$ —must satisfy $\psi(\tilde{m}) = \hat{m}$, following an inspection of (19). Consider three cases. If $\hat{m} \in (\underline{\beta}, \bar{\beta})$, then $\psi(\hat{m}) = \hat{m}$. In the equilibrium model, $\hat{m}$ corresponds to an SCE belief of productivity. If, on the other hand, $\hat{m} \leq \underline{\beta}$, then $\tilde{m} = \underline{\beta}$. Near the steady state, the m -dimension of the ODE changes at rate $\psi(\underline{\beta}) - m$, until the system settles at $\hat{m} = \psi(\underline{\beta})$. Therefore, $\psi(\underline{\beta}) \leq \underline{\beta}$ must hold, or equivalently $\tilde{\psi}(\underline{\beta}) = \underline{\beta}$. Correspondingly, $\underline{\beta}$ is a Berk–NE belief of productivity in the equilibrium model. The argument for the case $\hat{m} \geq \bar{\beta}$ is analogous.

The stability properties of Berk–Nash equilibria parallel those of the ODE's steady states. Since the second dimension of the ODE always converges, stability is determined solely by the first dimension. If $\hat{m} \in (\underline{\beta}, \bar{\beta})$ and ψ crosses the 45-degree line strictly from above at $\hat{m}$, then the steady state $(\hat{m}, \hat{\xi})$ is a sink and so is locally asymptotically stable in the Lyapunov sense. If, on the other hand, the crossing occurs strictly from below, then the steady state is a saddle point and is therefore Lyapunov unstable. If $\hat{m} \leq \underline{\beta}$, the steady state is again a sink, following the description in the above paragraph. The case $\hat{m} \geq \bar{\beta}$ is analogous.

By Theorem 1, steady states, ordered by their m -dimension from largest to smallest, must alternate in type: sink $\rightarrow$ saddle $\rightarrow$ sink $\rightarrow \dots$. Moreover, the m -coordinate of a saddle point must lie in the open interval $(\underline{\beta}, \bar{\beta})$, and its attraction basin is the vertical line passing through it. Every other point in Ξ lies in the attraction basin of a sink.

Figure 3 illustrates the above discussions in the context of Applications 1 and 2.

Convergence to sinks. We establish Theorem 4 for the uniform prior.

Theorem 5. *The process in (18) converges a.s. to the sinks of (19), and to every sink with positive probability.*

The proof of Theorem 5 involves multiple sizable building blocks, which we collect in Appendix C.4. Here we outline the proof and emphasize the role played each building block.

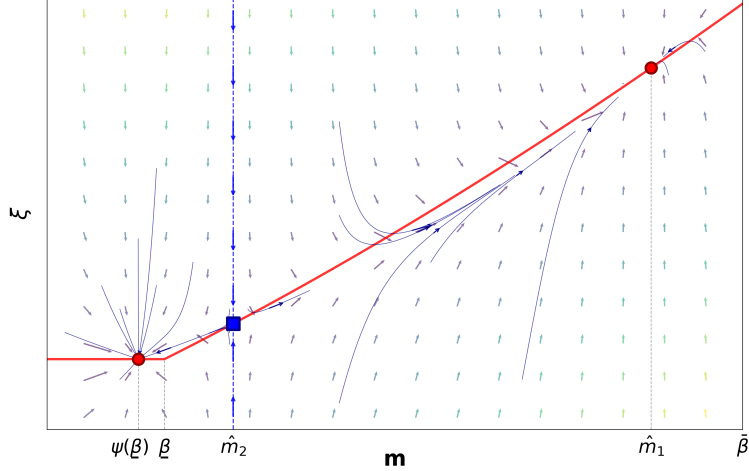


Figure 3: Linear-quadratic example for a small $\Delta > 0$: the red curve separates the regions in which $\dot{\xi}$ is positive and negative, respectively. The blue square is a saddle point, whose attraction basin is the blue dashed line. Every other point lies in the attraction basin of a sink, marked as a red dot.

Setting initial values as $\theta_0 = 0$, $F(\theta_0) = 0$, and $h_1(\theta_0) = h_1$, the update equation (18) can be expressed as

$$\theta_{n+1} - \theta_n = \underbrace{\gamma_{n+1}}_{=\frac{1}{n+1}} \left(F(\theta_n) + \underbrace{U_{n+1}}_{=b_{n+1}(\theta_n)\epsilon_{n+1}} + \underbrace{d_{n+1}}_{=F_{n+1}(\theta_n) - F(\theta_n)} \right), \quad n \in \mathbb{N}_0. \quad (20)$$

In stochastic approximation, γ_{n+1} is referred to as the *step size*, U_{n+1} as a *martingale-difference noise*, and d_{n+1} as an *approximation error*. Both γ_{n+1} and d_{n+1} are adapted to the public history H_n , while $\mathbb{E}[U_{n+1} \mid H_n] = 0$.

In our setting, $\gamma_{n+1} = 1/(n+1)$ holds because learning is Bayesian. $\{U_{n+1}\}$ arises from A's innate ability and performance errors; by construction, the process is subgaussian. Finally, $\{d_{n+1}\}$ captures the error introduced when the ODE replaces the true truncated-normal posterior with a degenerate belief. This approximation changes E's assessment choice and, in turn, A's effort choice relative to the true Bayesian benchmarks.

We first demonstrate that starting from a large n , the sample path of the state $\{\theta_{n+i}\}_{i=1}^k$ over the next k periods— k small relative to n —stays arbitrarily close to the trajectory of the ODE starting from θ_n . This property, referred to as *shadowing* in stochastic approximation, allows us to approximate the limiting behavior of $\{\theta_n\}$ by the ODE's trajectories.

Given shadowing, we apply Theorem 5.7 of B99, “limit-set theorem.” In the current environment, which is effectively one-dimensional, the theorem implies a.s. convergence of $\{\theta_n\}$ to the set of steady states of the ODE.

We next establish that convergence to any saddle point occurs with zero probability. Intuitively, this is because the attraction basin of a saddle point is a vertical line, while shocks to the state arising from the random outcomes come in the horizontal direction. After many shocks, the state escapes a small neighborhood around the saddle point and does not return with positive probability. Formally, we apply Proposition 5 of MY23, verifying that its conditions are satisfied in our setting.

Combining the above arguments yields a.s. convergence to the set of sinks.

To establish positive-probability convergence to any given sink, we apply Theorem 7.3 of B99, the “attractor theorem.” Under the full-support assumption, θ_n belongs to a small neighborhood of the sink with positive probability for every $n \in \mathbb{N}$. Starting from there, the trajectory of the ODE is attracted to the sink by definition. For large n , the sample path is shadowed by the ODE’s trajectory, so it, too, converges to the sink.

The role of convergence rates. It is useful to note the roles played by the convergence rates in our proofs.

Our applications of B99’s limit-set theorem and attractor theorem rely on the martingale-difference noise process $\{U_{n+1}\}$ being subgaussian. Consequently, starting from a large n , the cumulative impact of such noises over any fixed time span, weighted by the step sizes, vanishes as n grows, at a rate that depends only on a few primitive objects. This last universality property is required for the attractor theorem to apply. Moreover, we show that $d_{n+1} \rightarrow 0$ a.s. as n grows, so the approximation error is asymptotically negligible. Together, these bounds ensure the applicability of B99.

By contrast, establishing zero-probability convergence to saddle points requires stronger rate assumptions. A classical reference is P90, which assumes bounded martingale-difference noise and a (possibly nondeterministic) step size that decays at a sufficiently fast polynomial rate as n grows. MY23 generalize P90 to Gaussian martingale-difference noise, while requiring approximation errors to decay at rate $O(1/n)$ near the target saddle point. In our setting, this last rate assumption holds under the uniform prior and the resulting truncated normal posteriors.

Under general priors, the Laplace approximation in Appendix C.1 introduces additional approximation errors that may slow the rate of convergence, a challenge we now turn to.

C.3 Stochastic approximation: General priors

This appendix extends earlier analysis in Appendix C.2 to general priors.

We begin by quantifying the prior’s influence on the updating equation, ignoring its impact on players’ strategic behavior. That is, we continue to hold $F_{n+1}(\cdot)$ and $b_{n+1}(\cdot)$ as

in (18). We focus on what happens from time 1 onward and so take $n \in \mathbb{N}$; what happens between time 0 and 1 is clear.

Rewrite (13) as

$$m_{n+1} - m_n = \frac{1}{n+1} \left(F_{n+1,1}(\theta_n) + b_{n+1,1}(\theta_n) \epsilon_{n+1} \right) + \frac{1}{\sum_{i=1}^{n+1} I_i} \frac{d \log \bar{p}_0}{d\beta} \Big|_{\beta=m_{n+1}} - \frac{1}{\sum_{i=1}^n I_i} \frac{d \log \bar{p}_0}{d\beta} \Big|_{\beta=m_n},$$

where the subscript “1” indicates the first coordinate, the first line on the right-hand side also appeared in (17), and the second line represents the influence of the prior on the posterior-mode update. In Appendix C.5, we demonstrate that $\{\theta_n\}$ is a.s. bounded. Thus a.s. we have $|F_{n+1,1}(\theta_n)|, |b_{n+1,1}(\theta_n)| = O(1)$, and moreover

$$\begin{aligned} & \left| \frac{1}{\sum_{i=1}^{n+1} I_i} \frac{d \log \bar{p}_0}{d\beta} \Big|_{\beta=m_{n+1}} - \frac{1}{\sum_{i=1}^n I_i} \frac{d \log \bar{p}_0}{d\beta} \Big|_{\beta=m_n} \right| \\ &= \frac{1}{\sum_{i=1}^{n+1} I_i} \left| \frac{d \log \bar{p}_0}{d\beta} \Big|_{\beta=m_{n+1}} - \frac{d \log \bar{p}_0}{d\beta} \Big|_{\beta=m_n} - \frac{I_{n+1}}{\sum_{i=1}^n I_i} \frac{d \log \bar{p}_0}{d\beta} \Big|_{\beta=m_n} \right| \\ &\leq \frac{1}{(n+1)} \left| \text{const} \cdot (m_{n+1} - m_n) + O\left(\frac{1}{n}\right) \right|, \end{aligned}$$

where the constant in the last line depends only on m_n , m_{n+1} , $\underline{I}$, $\bar{I}$, and $\bar{p}_0$. Meanwhile, recall that ϵ_n ’s are standard normal, so

$$\mathbb{P}(|\epsilon_n| > 2\sqrt{\log n}) \leq \frac{2}{n^2}.$$

Since the right-hand side is summable, applying the Borel–Cantelli lemma yields

$$|\epsilon_n| \leq 2\sqrt{\log n} \quad \text{a.s. for large } n.$$

Combining these bounds yields

$$\begin{aligned} |m_{n+1} - m_n| &\leq \frac{1}{n+1} \left(|F_{n+1,1}(\theta_n)| + |b_{n+1,1}(\theta_n) \epsilon_{n+1}| + \text{const} \cdot |m_{n+1} - m_n| + O\left(\frac{1}{n}\right) \right) \\ &\implies |m_{n+1} - m_n| = O\left(\frac{\sqrt{\log n}}{n}\right) \quad \text{a.s.} \end{aligned}$$

where the last bound is determined by the term $b_{n+1,1}(\theta_n) \epsilon_{n+1}$. Substituting it into the upper

bound for $\left| \frac{1}{\sum_{i=1}^{n+1} I_i} \frac{d \log \bar{p}_0}{d\beta} \Big|_{\beta=m_{n+1}} - \frac{1}{\sum_{i=1}^n I_i} \frac{d \log \bar{p}_0}{d\beta} \Big|_{\beta=m_n} \right|$ yields

$$m_{n+1} - m_n = \frac{1}{n+1} \left(F_{n+1,1}(\theta_n) + b_{n+1,1}(\theta_n) \epsilon_{n+1} + O\left(\frac{\sqrt{\log n}}{n}\right) \right) \quad \text{a.s.}$$

A similar derivation applies to ξ . Rewrite (14) as

$$\xi_{n+1} - \xi_n = \frac{1}{n+1} (I_{n+1} - \xi_n) - \left(\frac{1}{n+1} \frac{d^2 \log \bar{p}_0}{d\beta^2} \Big|_{\beta=m_{n+1}} - \frac{1}{n} \frac{d^2 \log \bar{p}_0}{d\beta^2} \Big|_{\beta=m_n} \right).$$

Substituting in the bound for $|m_{n+1} - m_n|$ yields

$$\xi_{n+1} - \xi_n = \frac{1}{n+1} \left(I_{n+1} - \xi_n + O\left(\frac{\sqrt{\log n}}{n}\right) \right) \quad \text{a.s.}$$

In vector form, the update equation becomes

$$\theta_{n+1} - \theta_n = \frac{1}{n+1} \left(F_{n+1}(\theta_n) + b_{n+1}(\theta_n) \epsilon_n + O\left(\frac{\sqrt{\log n}}{n}\right) \right) \quad \text{a.s.} \quad (21)$$

Compared to the uniform-prior case, now there is a new approximation error of $O(\sqrt{\log n}/n)$, which, as shown in Appendix C.5, still vanishes sufficiently fast to ensure zero-probability convergence to every saddle point. Roughly speaking, this is because the logarithm is sub-polynomial. Thus, those bounds in the derivations of P90 and MY23 affected by the additional error—which require a polynomial rate of convergence—continue to hold.

Certainly, the prior also influences players’ strategic behavior. Had we calculated the choices of assessment and effort using the true Bayesian posterior, rather than the (Laplace) approximated, truncated normal posterior, the results would be slightly different. In Appendix C.5, we treat this difference as another approximation error rather than solving it explicitly. After the relevant adjustment, the total approximation error remains of $O(\sqrt{\log n}/n)$, so everything said earlier continues to hold.

Indeed, now with the freedom to choose the prior, we can concentrate it around a sink, and appeal to Benaïm’s (1999) “attractor theorem” to show that converge to this sink occurs with probability arbitrarily close to one.

C.4 Proof of Theorem 5

We follow the notation convention developed in B99, supplementing it with extensive use of big- O notation of the form $O(\text{function}(n))$, with the understanding that the function, e.g.,

$1/n$, may be undefined at $n = 0$. In such cases, we set the big- O term equal to zero (or any constant) at $n = 0$; such choice is irrelevant for the asymptotics.

Recall the update equation:

$$\theta_{n+1} - \theta_n = \underbrace{\gamma_{n+1}}_{=\frac{1}{n+1}} \left(F(\theta_n) + \underbrace{U_{n+1}}_{=b_{n+1}(\theta_n)\epsilon_{n+1}} + \underbrace{d_{n+1}}_{=F_{n+1}(\theta_n) - F(\theta_n)} \right), \quad n \in \mathbb{N}_0$$

with initial values $\theta_0 = 0$, $F(\theta_0) = 0$, and $h_1(\theta_0) = h_1$. The following conditions are required to apply the existing results in stochastic approximation to our setting.

Condition 1. $\sum_{n=0}^{\infty} \gamma_{n+1} = \infty$ and $\sum_{n=0}^{\infty} \gamma_{n+1}^2 < \infty$.

Condition 2. $F : \Xi \rightarrow \mathbb{R}^2$, $\theta \mapsto F(\theta)$ is Lipschitz.

Condition 3. $\{U_{n+1}\}$ is subgaussian with $\mathbb{E}[\|U_{n+1}\|^2] \in [\underline{\Gamma}, \bar{\Gamma}]$, where $\underline{\Gamma} := \underline{I}/(\bar{I})^2$ and $\bar{\Gamma} := \bar{I}/(\underline{I})^2$ are both positive.

Condition 4. $\{\theta_n\}$ is a.s. bounded.

Condition 5. $d_{n+1} = F_{n+1}(\theta_n) - F(\theta_n) \rightarrow 0$ as $n \rightarrow \infty$ a.s.

Condition 6. For any saddle point $\hat{\theta}$ of the ODE in (19), there exists a neighborhood U of $\hat{\theta}$ such that $F_n(\theta) - F(\theta) = O(1/n)$ across $\theta \in U$ and $n \in \mathbb{N}$.

Condition 7. For a saddle point $\hat{\theta}$ and Jacobian $\nabla F|_{\theta=\hat{\theta}}$, identify eigenvalues with negative real parts, and denote the affine space spanned by the corresponding eigenvectors by H . There exist constant $C > 0$ and a neighborhood U of $\hat{\theta}$ such that for sufficiently large n , $|b_n(\theta) \cdot h| \geq C$ holds for all $h \in H^*$ and $\theta \in U$, where H^* denotes the set of all unit vectors orthogonal to H .

Conditions 1–3 are easy to verify. The proofs of Conditions 4–6 build on the next lemma.

Lemma 1. For $n \in \mathbb{N}$:

- (i) $h_n(\theta) - h(\tilde{m}) = O(1/\sqrt{n})$ uniformly over $\theta \in \Xi$.
- (ii) For each $m \in (\underline{\beta}, \bar{\beta})$, there exists a neighborhood of U of m such that the bound in Part (i) can be improved to $O(1/n)$ across $\theta \in U \times [\underline{I}, \bar{I}]$.

Proof. We state a useful fact regarding truncated normal distributions (see, e.g., Lemma 1 of MY23):

Fact 2. For $n \in \mathbb{N}$ and $\theta = (m, \xi) \in \Xi$, denote the mean of $\tilde{\mathcal{N}}(m, 1/(n\xi))$ by $\nu_n(\theta)$. There exists a constant $C > 0$ such that

$$|\nu_n(\theta) - \tilde{m}| < \frac{C}{\sqrt{n}} \quad \text{for all } n \in \mathbb{N} \text{ and } \theta \in \Xi.$$

Moreover, for each $m' \in (\underline{\beta}, \bar{\beta})$, there exists a neighborhood U' of m' and a constant $C' > 0$ (which depends on U') such that

$$|\nu_n(\theta) - m| < \frac{C'}{n} \quad \text{for all } n \in \mathbb{N} \text{ and } \theta \in U' \times [\underline{I}, \bar{I}].$$

We now prove Lemma 1. Fix $\theta = (m, \xi) \in \Xi$. For large n , we seek to bound

$$\left| \mathbb{E}_{\beta \sim \tilde{\mathcal{N}}(m, 1/(n\xi))} \left[\frac{\partial V_E}{\partial h}(h, \beta) \right] - \frac{\partial V_E}{\partial h}(h, \tilde{m}_n) \right|.$$

A second-order Taylor expansion of $\frac{\partial V_E}{\partial h}(h, \beta)$ around $\beta = \nu_n(\theta)$ yields

$$\frac{\partial V_E}{\partial h}(h, \beta) - \frac{\partial V_E}{\partial h}(h, \nu_n(\theta)) = \frac{\partial^2 V_E}{\partial h \partial \beta}(h, \nu_n(\theta)) (\beta - \nu_n(\theta)) + \frac{1}{2} \frac{\partial^3 V_E}{\partial h \partial \beta^2}(h, z(\beta)) (\beta - \nu_n(\theta))^2,$$

where $z(\beta)$ lies between β and $\nu_n(\theta)$ and thus within $[\underline{\beta}, \bar{\beta}]$. Taking expectation of the right-hand side over $\beta \sim \tilde{\mathcal{N}}(m, 1/(n\xi))$ yields

$$0 + O(1) \cdot \frac{1}{n\underline{I}} = O\left(\frac{1}{n}\right),$$

where the zero term obtains because $\mathbb{E}[\beta] = \nu_n(\theta)$, and the big O term follows from the smoothness of V_E and the variance inequality $\mathbb{V}(\beta) \leq 1/(n\xi) \leq 1/(n\underline{I})$. Therefore,

$$\mathbb{E}_{\beta \sim \tilde{\mathcal{N}}(m, 1/(n\xi))} \left[\frac{\partial V_E}{\partial h}(h, \beta) \right] = \frac{\partial V_E}{\partial h}(h, \nu_n(\theta)) + O\left(\frac{1}{n}\right),$$

and consequently

$$\begin{aligned} & \left| \mathbb{E}_{\beta \sim \tilde{\mathcal{N}}(m, 1/(n\xi))} \left[\frac{\partial V_E}{\partial h}(h, \beta) \right] - \frac{\partial V_E}{\partial h}(h, \tilde{m}) \right| \\ & \leq \left| \mathbb{E}_{\beta \sim \tilde{\mathcal{N}}(m, 1/(n\xi))} \left[\frac{\partial V_E}{\partial h}(h, \beta) \right] - \frac{\partial V_E}{\partial h}(h, \nu_n(\theta)) \right| + \left| \frac{\partial V_E}{\partial h}(h, \nu_n(\theta)) - \frac{\partial V_E}{\partial h}(h, \tilde{m}) \right| \\ & = O\left(\frac{1}{n}\right) + O(|\nu_n(\theta) - \tilde{m}|). \end{aligned}$$

By Fact 2, the last line is $O(1/\sqrt{n})$ uniformly over Ξ , and $O(1/n)$ in a neighborhood of θ if $m \in (\underline{\beta}, \bar{\beta})$.

Next, recall that $h_{n+1}(\theta)$ and $h(\tilde{m})$ are determined by the first-order conditions

$$\mathbb{E}_{\beta \sim \tilde{\mathcal{N}}(m, 1/(n\xi))} \left[\frac{\partial V_E}{\partial h}(h, \beta) \right] = \kappa'(h) \quad \text{and} \quad \frac{\partial V_E}{\partial h}(h, \tilde{m}) = \kappa'(h), \quad \text{respectively.}$$

Since V_E and κ are both smooth functions, the bounds established above pass to $h_{n+1}(\theta) - h(\tilde{m})$. $\square$

Proof of Conditions 4–6. Since $\{\xi_n\}$ is bounded by definition, it suffices to show that $\{m_n\}$ is a.s. bounded. Denote $C_{n+1} := n\xi_n/I_{n+1}$ and rewrite (17) as

$$m_{n+1} = \frac{C_{n+1}m_n + \psi(\tilde{m}_n)}{1 + C_{n+1}} + \frac{U_{n+1}}{n+1} + \frac{(\beta \circ h_{n+1})(\theta_n) - (\beta \circ h)(\tilde{m}_n)}{1 + C_{n+1}}.$$

Since $C_{n+1} = \Theta(n)$ and $h_{n+1}(\theta_n) - h(\tilde{m}_n) = O(1/\sqrt{n})$, the third term on the right-hand side is of $o(1/n)$. Omitting this negligible term for ease of notation, we obtain

$$m_{n+1} \leq \frac{C_{n+1}m_n + \bar{\psi}}{1 + C_{n+1}} + \frac{U_{n+1}}{n+1}, \quad \text{where } \bar{\psi} = \sup_{\beta \in [\underline{\beta}, \bar{\beta}]} \psi.$$

Consider a dominating process defined by

$$m'_{n+1} = \frac{C_{n+1}m'_n + \bar{\psi}}{1 + C_{n+1}} + \frac{U_{n+1}}{n+1}.$$

If $m_n \leq m'_n$, then

$$m_{n+1} \leq \frac{C_{n+1}m_n + \bar{\psi}}{1 + C_{n+1}} + \frac{U_{n+1}}{n+1} \leq \frac{C_{n+1}m'_n + \bar{\psi}}{1 + C_{n+1}} + \frac{U_{n+1}}{n+1} = m'_{n+1}.$$

Applying this argument iteratively shows that for all $k \geq 1$,

$$\begin{aligned} m_{n+k} &\leq m'_{n+k} \\ &= \left(\prod_{j=1}^k \frac{C_{n+j}}{1 + C_{n+j}} \right) m'_n + \left(1 - \left(\prod_{j=1}^k \frac{C_{n+j}}{1 + C_{n+j}} \right) \right) \bar{\psi} + O_p \left(\sqrt{\sum_{j=1}^k \frac{1}{(n+j)^2}} \right) \\ &\leq \max\{m'_n, \bar{\psi}\} + O_p \left(\sqrt{\sum_{j=1}^k \frac{1}{(n+j)^2}} \right), \end{aligned}$$

where the equality holds up to $O_p\left(\sqrt{\sum_{j=1}^k \frac{1}{(n+j)^2}}\right)$ by Condition 3. Since the term $\sqrt{\cdot}$ remains finite as $k \rightarrow \infty$, the last line is a.s. bounded from above by Kolmogorov's convergence criterion. A lower bound can be derived analogously, thus establishing Condition 4.

Conditions 5 and 6 follow immediately. $\square$

Proof of Condition 7. Recall that a saddle point $\hat{\theta} = (\hat{m}, \hat{\xi})$ must satisfy $\hat{m} \in (\underline{\beta}, \bar{\beta})$ and so $\psi(\hat{m}) = \hat{m}$, and moreover $\hat{\xi} = (I \circ h)(\hat{m})$. Linearizing F around $\hat{\theta}$ yields

$$\begin{aligned} & \nabla F|_{\theta=\hat{\theta}} \\ &= \left[\begin{array}{cc} \frac{(I \circ h)'(m) \cdot (\psi(m) - m) + (I \circ h)(m) \cdot (\psi'(m) - 1)}{\xi} & -\frac{(I \circ h)(m) \cdot (\psi(m) - m)}{\xi^2} \\ (I \circ h)'(m) & -1 \end{array} \right] \Big|_{\theta=\hat{\theta}} \\ &= \begin{bmatrix} \psi'(\hat{m}) - 1 & 0 \\ (I \circ h)'(\hat{m}) & -1 \end{bmatrix}. \end{aligned}$$

The eigenvalues of $\nabla F|_{\theta=\hat{\theta}}$ are its diagonal entries: $\psi'(\hat{m}) - 1$ and -1 . The first is positive because ψ crosses the 45 degree line from below at $\hat{m}$. The second is negative and corresponds to an eigenvector of the form $[0, y]^\top$. It follows that $H^* = \{[1, 0]^\top\}$, and moreover

$$b_n(\theta) \cdot [1, 0]^\top = b_{n,1}(\theta) \geq \sqrt{\underline{\Gamma}} > 0 \text{ for all } \theta \in \Xi \text{ and } n \in \mathbb{N}. \quad \square$$

Proof of Theorem 5. The proof was outlined in Appendix C.2. Here, we provide the omitted details.

Preliminaries. The following concepts and notation are taken from B99.

Define $\tau_0 = 0$ and $\tau_n = \sum_{i=1}^n \gamma_i$ for $n \in \mathbb{N}$. The inverse of the map $n \rightarrow \tau_n$ is defined by

$$\tau^{-1}(t) = \sup\{k \geq 0 : t \geq \tau_k\}, \quad t \geq 0.$$

By Condition 1, $\lim_{t \rightarrow \infty} \tau^{-1}(t) = \infty$.

Define the *continuous-time affine-interpolated process* $\theta : \mathbb{R}_+ \rightarrow \Xi$ by

$$\theta(\tau_n + s) = \theta_n + s \frac{\theta_{n+1} - \theta_n}{\tau_{n+1} - \tau_n}, \quad n \in \mathbb{N}_0 \text{ and } 0 \leq s < \gamma_{n+1}.$$

The *limit set* of this process is given by

$$L(\theta) = \bigcap_{t \geq 0} \overline{\theta([t, \infty))}.$$

A point $a \in \Xi$ is *attainable* by this process if for all $t > 0$ and neighborhood U of a ,

$$\mathbb{P}(\exists s \geq t : \theta(s) \in U) > 0.$$

The set of attainable points is denoted by $Att(\theta)$.

Let $\Phi : \mathbb{R}_+ \times \Xi \rightarrow \Xi$, $(t, \theta) \mapsto \Phi(t, \theta) = \Phi_t(\theta)$, denote the *semiflow* induced by F , such that

$$\Phi_l(\theta(t)) = \theta(t) + \int_t^{t+l} F(\theta(s)) ds \quad \text{for all } (t, l) \in \mathbb{R}_+^2.$$

$\theta : \mathbb{R}_+ \rightarrow \Xi$ is an *asymptotic pseudotrajectory* for Φ if

$$\lim_{t \rightarrow \infty} \sup_{0 \leq l \leq T} \|\theta(t+l), \Phi_l(\theta(t))\| = 0 \quad \text{for all } T > 0.$$

This means that for sufficiently large t , the curve $[0, T] \rightarrow \Xi$, $l \mapsto \theta(t+l)$ shadows the Φ -trajectory of the point $\theta(t)$ over the interval $[0, T]$ with arbitrary accuracy.

We wish to demonstrate that $\theta : \mathbb{R}_+ \rightarrow \Xi$ is an asymptotic pseudotrajectory of Φ . To this end, define two continuous time processes $\bar{U} : \mathbb{R}_+ \rightarrow \mathbb{R}^2$ and $\bar{\gamma} : \mathbb{R}_+ \rightarrow \mathbb{R}$ such that

$$\bar{U}(\tau_n + s) = U_{n+1}, \quad \bar{\gamma}(\tau_n + s) = \gamma_{n+1}$$

for all $n \in \mathbb{N}_0$ and $0 \leq s < \gamma_{n+1}$. Define

$$\Delta(t, T) = \left\| \sup_{0 \leq l \leq T} \int_t^{t+l} \bar{U}(s) ds \right\|$$

for $t, T \geq 0$. Intuitively, $\Delta(t, T)$ represents the cumulative martingale-difference noises, weighted by step sizes, over $[t, t+T]$.

A point $a \in \Xi$ is *chain recurrent* if for all $\delta, T > 0$, there exists a finite sequence of partial trajectories

$$\{\Phi_t(y_i) : 0 \leq t \leq t_i\}; \quad i = 0, \dots, k-1; \quad t_i \geq T$$

such that

$$\begin{aligned} \|y_0 - a\| &< \delta; \\ \|\Phi_{t_j}(y_j) - y_{j+1}\| &< \delta, \quad j = 0, \dots, k-1; \\ y_k &= a. \end{aligned}$$

For a nonempty invariant set Ξ' of Φ , we say that Φ is *chain recurrent on Ξ'* if every point in Ξ' is chain recurrent for Φ restricted to Ξ' . A compact invariant set on which Φ is chain recurrent is called *internally chain recurrent*.

In our setting, every invariant set of Φ is included in the set of steady states. Since the latter is finite and chain recurrent (by definition), the three concepts coincide: invariant points, internally chain recurrent points, and steady states.

Step 1. We demonstrate that $\{\theta_n\}$ a.s. converges to the steady states of Φ . The proof appeals to Proposition 4.1 of B99, which requires three conditions:

$$\text{A1 } \lim_{t \rightarrow \infty} \Delta(t, T) = 0 \text{ for all } T > 0.$$

$$\text{A2 } \sup_n \|\theta_n\| < \infty.$$

$$\text{A2'} } F \text{ is Lipschitz and bounded on a neighborhood of } \{\theta_n\}.$$

Proposition 4.4 and Remark 4.5 of B99 show that under Conditions 1, 3, and 5, A1 holds a.s. Indeed, for all $\alpha, t, T > 0$,

$$\mathbb{P}(\Delta(t, T) \geq \alpha) \leq C \exp \left(-\frac{\alpha^2}{C' T \bar{\gamma}(u)} \right) \quad (22)$$

holds for some $t \leq u \leq t + T$, and some positive constants C, C' that depend only on $\dim(\Xi)$ ($= 2$) and $\bar{\Gamma}$ (the sup subgaussian coefficient). Meanwhile, A2 holds a.s. by Condition 4, and A2' holds a.s. by Conditions 2 and 4. Proposition 4.1 of B99 states that under A1 and either A2 or A2', $\theta : \mathbb{R}_+ \rightarrow \Xi$ is an asymptotic pseudotrajectory of Φ . Moreover, under A2',

$$\sup_{0 \leq l \leq T} \|\theta(t + l) - \Phi_l(\theta(t))\| \leq C(T) [\Delta(t - 1, T + 1) + \sup_{t \leq s \leq t + T} \bar{\gamma}(s)] \quad (23)$$

holds for sufficiently large t , where $C(T)$ is a positive constant that depends only on T and F .

Theorem 5.7 of B99, the “limit-set theorem”, states that if, in addition, $\theta : \mathbb{R}_+ \mapsto \Xi$ is precompact, i.e., its image has a compact closure in Ξ , then its limit set $L(\theta)$ is internally chain recurrent. In our setting, this additional requirement is satisfied a.s. by Condition 4. Since internally chain recurrent points and steady states are equivalent in our setting (established in the preliminaries), $L(\theta)$ is a.s. included in the set of the steady states of Φ .

Step 2. We demonstrate that $\{\theta_n\}$ converges to any saddle point $\hat{\theta}$ of Φ with zero probability. We appeal to Proposition 5 of MY23, having verified its required conditions 1–7.

Step 3. We demonstrate that $\{\theta_n\}$ converges to every sink of Φ with positive probability. Let $\{\mathcal{F}_t : t \geq 0\}$ denote the filtration generated by $\theta : \mathbb{R}_+ \rightarrow \Xi$. Fix $\delta, T > 0$. For sufficiently large t ,

$$\begin{aligned}
& \mathbb{P} \left(\sup_{0 \leq l \leq T} \|\theta(t+l) - \Phi_l(\theta(t))\| \geq \delta \mid \mathcal{F}_t \right) \\
& \leq \mathbb{P} \left(C(T)[\Delta(t-1, T+1) + \sup_{t \leq s \leq t+T} \bar{\gamma}(s)] \geq \delta \right) \quad (\because (23)) \\
& = \mathbb{P} \left(\Delta(t-1, T+1) \geq \frac{\delta}{C(T)} - \bar{\gamma}(t) \right) \quad (\because \bar{\gamma}(t) \text{ is decreasing in } t) \\
& \leq \sup_{t \leq u \leq t+T} C \exp \left(-\frac{(\delta/C(T) - \bar{\gamma}(t))^2}{C'T\bar{\gamma}(u)} \right) \quad (\because (22)) \\
& = C \exp \left(-\frac{(\delta/C(T) - \bar{\gamma}(t))^2}{C'T\bar{\gamma}(t)} \right) \quad (\because \bar{\gamma}(t) \text{ is decreasing in } t) \\
& := w(t, \delta, T).
\end{aligned}$$

As $t \rightarrow \infty$, $\bar{\gamma}(t)$ vanishes. Therefore the upper bound $w(t, \delta, T)$ vanishes, verifying Condition (24) of B99.

A sink $\hat{\theta}$ of Φ is, by definition, a point attractor. Denote its attraction basin by $B(\hat{\theta})$, clearly $\hat{\theta} \in B(\hat{\theta})$. Let $N_\epsilon(\hat{\theta})$ denote the ϵ -neighborhood of $\hat{\theta}$ in Ξ , $N_\epsilon(\hat{\theta}) \subset B(\hat{\theta})$ holds for sufficiently small ϵ . By the full-support assumption, $\mathbb{P}(\theta_n \in N_{\epsilon/2}(\hat{\theta})) > 0$ for $n \in \mathbb{N}$. Starting from a sufficiently large $t = \tau^{-1}(n)$ such that $\theta(t) = \theta_n \in N_{\epsilon/2}(\hat{\theta}) \subset B(\hat{\theta})$, the Φ -trajectory is attracted to $\hat{\theta}$:

$$\|\Phi_l(\theta(t)) - \hat{\theta}\| \leq \|\theta(t) - \hat{\theta}\| < \frac{\epsilon}{2} \text{ for all } l > 0.$$

Combined with (23), which implies

$$\sup_{0 \leq l \leq T} \|\theta(t+l) - \Phi_l(\theta(t))\| < \frac{\epsilon}{2}$$

for sufficiently large t , we obtain

$$\sup_{0 \leq l \leq T} \|\theta_{t+l} - \hat{\theta}\| \leq \sup_{0 \leq l \leq T} \|\theta_{t+l} - \Phi_l(\theta(t))\| + \sup_{0 \leq l \leq T} \|\Phi_l(\theta(t)) - \hat{\theta}\| \leq \frac{\epsilon}{2} + \frac{\epsilon}{2} = \epsilon,$$

and therefore

$$\mathbb{P}(\exists s \geq t : \theta(s) \in N_\epsilon(\hat{\theta})) > 0.$$

This establishes that $\hat{\theta}$ is attainable by $\theta : \mathbb{R}_+ \rightarrow \Xi$, i.e., $\hat{\theta} \in \text{Att}(\theta)$.

It follows that $\hat{\theta} \in B(\hat{\theta}) \cap \text{Att}(\theta)$. Theorem 7.3 of B99, the ‘‘attractor theorem’’, states that under Condition (24) of B99 (verified above), if $\text{Att}(\theta) \cap B(\hat{\theta}) \neq \emptyset$, then $\mathbb{P}(L(\theta) \subseteq \{\hat{\theta}\}) > 0$.

We therefore conclude that $\{\theta_n\}$ converges to $\hat{\theta}$ with positive probability. $\square$

C.5 Proof of Theorem 4

We provide the omitted details from Appendix C.3, proceeding in five steps.

Step 1. We quantify the strategic impact of a general prior. When deriving the update equation in Appendix C.3, we took E's assessment choice in period $n + 1$ as $h_{n+1}(\theta_n)$. This is the optimal assessment given the truncated normal posterior parameterized by (n, θ_n) , which differs from the optimal assessment given the true Bayesian posterior p_n . We treat this difference as an approximation error and bound its magnitude.

Suppose that $\{m_n\}$ is a.s. bounded—an assumption verified later. Then for large n , $m_n = O(1)$ and $v_n = \Theta(1/n)$ hold a.s. In Step 2 on the next page, we demonstrate that

$$\left| \mathbb{E}_{\beta \sim p_n} \left[\frac{\partial V_E}{\partial h}(h, \beta) \right] - \mathbb{E}_{\beta \sim \mathcal{N}(m_n, v_n)} \left[\frac{\partial V_E}{\partial h}(h, \beta) \right] \right| = o\left(\frac{1}{n}\right) \quad \text{a.s.}$$

Combined with the proof of Lemma 1, this implies

$$h_{n+1}(p_n) - h_{n+1}(\theta_n) = o\left(\frac{1}{n}\right) \quad \text{a.s.}$$

where $h_{n+1}(p_n)$ denotes the assessment choice given the true Bayesian posterior p_n . Rather than calculating $h_{n+1}(p_n)$ explicitly, we treat its difference from $h_{n+1}(\theta_n)$ as an approximation error, dominated by the O -terms we have previously encountered. Introducing this error and replicating the analysis line by line confirm that all previous derivations continue to hold in the big O sense. We therefore conclude that:

Lemma 2. *Under a general prior, the approximation error in the update equation (20) a.s. becomes*

$$d_{n+1} = F_{n+1}(\theta_n) - F(\theta_n) + O\left(\frac{\sqrt{\log n}}{n}\right)$$

where $F_{n+1}(\theta_n)$ retains its definition in (18).

It follows that Conditions 4 and 5 continue to hold (confirming that the assumption $\{m_n\}$ is a.s. bounded is without loss). Condition 6 weakens to the following:

Condition 8. *Fix any saddle point $\hat{\theta}$ of the ODE, and let U be the neighborhood of $\hat{\theta}$ in Lemma 1(ii). Across $\theta \in U$, $F_{n+1}(\theta) - F(\theta) + O(\sqrt{\log n}/n) = O(1/n) + O(\sqrt{\log n}/n) = O(\sqrt{\log n}/n)$ a.s.*

Finally, Conditions 1, 2, 3, and 7 continue to hold as they do not depend on the prior.

Step 2. We demonstrate that $|h_{n+1}(p_n) - h_{n+1}(\theta_n)| = o(1/n)$ a.s.

Consider the a.s. event in which $m_n = O(1)$ and $v_n = \Theta(1/n)$. By Laplace's approximation in Appendix C.1,

$$\mathbb{E}_{\beta \sim p_n} \left[\frac{\partial V_E}{\partial h} \right] - \mathbb{E}_{\beta \sim \tilde{\mathcal{N}}(m_n, v_n)} \left[\frac{\partial V_E}{\partial h} \right] = \frac{\text{Cov}_{\beta \sim \tilde{\mathcal{N}}(m_n, v_n)} \left(\frac{\partial V_E}{\partial h}(h, \beta), \exp(K(\beta, m_n)(\beta - m_n)^3) \right)}{\mathbb{E}_{\beta \sim \tilde{\mathcal{N}}(m_n, v_n)} [\exp(K(\beta, m_n)(\beta - m_n)^3)]}.$$

Because K is smooth, $\beta \in [\underline{\beta}, \bar{\beta}]$, and $m_n = O(1)$, there exists a constant C_0 such that

$$|K(\beta, m_n)(\beta - m_n)^3| \leq C_0$$

uniformly in β and n . Therefore,

$$\mathbb{E}_{\beta \sim \tilde{\mathcal{N}}(m_n, v_n)} [\exp(K(\beta, m_n)(\beta - m_n)^3)] \geq e^{-C_0}.$$

It therefore suffices to show that

$$\left| \text{Cov}_{\beta \sim \tilde{\mathcal{N}}(m_n, v_n)} \left(\frac{\partial V_E}{\partial h}(h, \beta), \exp(K(\beta, m_n)(\beta - m_n)^3) \right) \right| = O(v_n^2) = o\left(\frac{1}{n}\right).$$

where the second equality uses $v_n = \Theta(1/n)$.

Fix $\beta_0 \in [\underline{\beta}, \bar{\beta}]$. Since covariance is unchanged by subtracting constants,

$$\begin{aligned} & \text{Cov}_{\beta \sim \tilde{\mathcal{N}}(m_n, v_n)} \left(\frac{\partial V_E}{\partial h}(h, \beta), \exp(K(\beta, m_n)(\beta - m_n)^3) \right) \\ &= \mathbb{E}_{\beta \sim \tilde{\mathcal{N}}(m_n, v_n)} \left[\left(\frac{\partial V_E}{\partial h}(h, \beta) - \frac{\partial V_E}{\partial h}(h, \beta_0) \right) \right. \\ & \quad \times \left(\exp(K(\beta, m_n)(\beta - m_n)^3) - \exp(K(\beta_0, m_n)(\beta_0 - m_n)^3) \right) \Big] \\ & \quad - \mathbb{E}_{\beta \sim \tilde{\mathcal{N}}(m_n, v_n)} \left[\frac{\partial V_E}{\partial h}(h, \beta) - \frac{\partial V_E}{\partial h}(h, \beta_0) \right] \\ & \quad \times \mathbb{E}_{\beta \sim \tilde{\mathcal{N}}(m_n, v_n)} [\exp(K(\beta, m_n)(\beta - m_n)^3) - \exp(K(\beta_0, m_n)(\beta_0 - m_n)^3)]. \end{aligned}$$

Applying the triangle inequality,

$$\begin{aligned}
& \left| \text{Cov}_{\beta \sim \tilde{\mathcal{N}}(m_n, v_n)} \left(\frac{\partial V_E}{\partial h}(h, \beta), \exp(K(\beta, m_n)(\beta - m_n)^3) \right) \right| \\
& \leq \mathbb{E}_{\beta \sim \tilde{\mathcal{N}}(m_n, v_n)} \left[\left| \frac{\partial V_E}{\partial h}(h, \beta) - \frac{\partial V_E}{\partial h}(h, \beta_0) \right| \right. \\
& \quad \left. \times \left| \exp(K(\beta, m_n)(\beta - m_n)^3) - \exp(K(\beta_0, m_n)(\beta_0 - m_n)^3) \right| \right] \\
& + \mathbb{E}_{\beta \sim \tilde{\mathcal{N}}(m_n, v_n)} \left[\left| \frac{\partial V_E}{\partial h}(h, \beta) - \frac{\partial V_E}{\partial h}(h, \beta_0) \right| \right] \\
& \quad \times \mathbb{E}_{\beta \sim \tilde{\mathcal{N}}(m_n, v_n)} \left[\left| \exp(K(\beta, m_n)(\beta - m_n)^3) - \exp(K(\beta_0, m_n)(\beta_0 - m_n)^3) \right| \right]
\end{aligned}$$

Then, by the smoothness of $\partial V_E / \partial h$ and K , the last line is bounded above by

$$C_1 \{ \mathbb{E}[|\beta - \beta_0|^4] + \mathbb{E}[|\beta - \beta_0|] \mathbb{E}[|\beta - \beta_0|^3] \}$$

for some positive constant C_1 .

The remainder of the proof distinguishes between two cases.

Case 1: $m_n \in (\underline{\beta}, \bar{\beta})$. In this case,

$$\mathbb{E}[|\beta - m_n|^j] = O(v_n^{j/2}), \quad j = 1, 2, 3, 4.$$

Setting $\beta_0 = m_n$ in the last upper bound gives

$$\mathbb{E}[|\beta - \beta_0|^4] + \mathbb{E}[|\beta - \beta_0|] \mathbb{E}[|\beta - \beta_0|^3] = O(v_n^2)$$

as claimed.

Case 2: $m_n \notin (\underline{\beta}, \bar{\beta})$. We focus on the case where $m_n \leq \underline{\beta}$; the case $m_n \geq \bar{\beta}$ is analogous. Using the moments bounds for a truncated normal distribution derived in [Horrace \(2015\)](#),

$$\mathbb{E}[(\beta - \underline{\beta})^j] = O \left(\left(\frac{v_n}{\underline{\beta} - m_n + \sqrt{v_n}} \right)^j \right), \quad j = 1, 2, 3, 4.$$

Since $v_n = \Theta(1/n)$, the right-hand side is again of $O(v_n^{j/2})$ when n is large. Setting $\beta_0 = \underline{\beta}$ then gives the desired result.

Step 3. Since Conditions 1–5 remain true, applying B99’s limit-set theorem yields a.s. convergence to the set of steady states of the ODE.

Step 4. We establish zero-probability convergence to any saddle point $\hat{\theta}$ by modifying the proof of MY23 Proposition 5. MY23's proof uses Condition 6: $F_{n+1}(\theta) - F(\theta) = O(1/n)$ across a neighborhood U of $\hat{\theta}$. In our setting, the rate of decay is $O(\sqrt{\log n}/n)$; recall Condition 8. This change affects Condition (32) of MY23: the term $O(S_n/n^2)$ on the right-hand side becomes $O(S_n\sqrt{\log n}/n^2)$. Everything else—including Condition (33) of MY23—remains unchanged.

Essentially, Conditions (32) and (33) of MY23, after our adaptation, can be expressed as

$$A_n \geq \frac{k_1 S_n^2}{n+1} + O\left(\frac{S_n \sqrt{\log n}}{n^2}\right), \quad B_n \geq \frac{k_2}{n^2},$$

where S_n and the constants k_1, k_2 are all positive. Based on these, we wish demonstrate that

$$A_n + B_n \geq \frac{\text{const}}{n^2}$$

for some positive constant. This is the key inequality on p. 709 of P90, or the first display on p. 16 of MY23's Online Appendix.

Our proof modifies the argument on p. 709 of P90, using the property that $\log n$ is subpolynomial. Specifically, fix $\gamma \in (0, 1)$ and $C > 0$. If $S_n > C/n^\gamma$, then the lower bound for A_n is asymptotically dominated by the term $\frac{k_1 S_n^2}{n+1} > 0$. It follows that asymptotically, $A_n > 0$, and so $A_n + B_n \geq k_2/n^2$. If, on the other hand, $S_n \leq C/n^\gamma$, then the lower bound for A_n may be dominated by the term $O(\frac{S_n \sqrt{\log n}}{n^2})$. Whenever this is the case, the following holds:

$$O\left(\frac{S_n \sqrt{\log n}}{n^2}\right) = o(1/n^2) \ll B_n,$$

so again $A_n + B_n \geq \frac{\text{const}}{n^2}$ asymptotically.

Step 5. We demonstrate that for a given sink $\hat{\theta}$, convergence occurs with probability arbitrarily close to one given a sufficiently concentrated prior around $\hat{\theta}$. Again, we apply B99's attractor theorem (Theorem 7.3), which states that for any open set U with compact closure contained in $B(\hat{\theta})$, there exist numbers $\delta(U), T(U) > 0$ depending on U (these are derived in Lemma 6.8 of B99) such that

$$\mathbb{P}(L(\theta) \subseteq \{\hat{\theta}\}) \geq [1 - w(t, \delta(U), T(U))] \mathbb{P}(\exists s \geq t : \theta(s) \in U).$$

As derived earlier,

$$w(t, \delta, T) = C \exp\left(-\frac{(\delta/C(T) - \bar{\gamma}(t))^2}{C'T\bar{\gamma}(t)}\right),$$

where C and C' depend only on $\dim(M) = 2$ and $\bar{\Gamma}$, whereas $C(T)$ depends only on T and F . Also noted earlier was that taking $t \rightarrow \infty$, holding everything else, sends $w(t, \delta, T) \rightarrow 0$. Now set $U = N_\epsilon(\hat{\theta})$, and let $\delta(U)$ and $T(U)$ be determined as in Lemma 6.8 of B99. Taking $\epsilon \rightarrow 0$ and the prior to be increasingly concentrated around $\hat{\theta}$, the lower bound for $\mathbb{P}(L(\theta) \subseteq \{\hat{\theta}\})$ tends to 1. $\square$

C.6 Discussion of Assumption 6

Assumption 6 is not innocuous when viewed globally. However, viewed locally, it is nothing but a first-order Taylor expansion of $R(h, \beta)$ near a fixed β_0 . This observation motivates an alternative approach we pursue in a companion project, which aims to dispense with Assumption 6 altogether: begin with a prior concentrated around a locally asymptotically stable steady state, and expand $R(h, \beta)$ linearly in β near that steady state. Over time, the posterior diverges from the prior. However, using posterior concentration results in Bayesian statistics (e.g., [van der Vaart and van Zanten 2008](#)), we demonstrate that with high probability, posterior beliefs remain within an $O(1/\sqrt{n})$ neighborhood of the steady state as $n \rightarrow \infty$. Within this neighborhood, the linear approximation remains valid. Applying B99's attractor theorem then yields positive-probability convergence to the steady state.

Compared to the approach developed in this paper, this alternative approach yields only positive-probability convergence to a locally asymptotically stable point, rather than a.s. convergence to the set of locally asymptotically stable points. In other words, it establishes local rather than global convergence. Its advantage is that it avoids imposing additional assumptions on the primitives. Outside one-dimensional environments, the limit set of the flow may contain complex objects such as cycles or strange attractors. In such environments, local convergence may be the best one can hope for, absent strong geometric conditions to pin down the limit set, so the advantage of the alternative approach becomes more significant.

Given the scope of this paper, we leave the full development of the alternative approach to a companion project.

D Learning under affirmative action

In this appendix, we examine the Bayesian learning foundation of the Berk-Nash equilibrium that arises under affirmative action.

There are $J \geq 2$ groups of agents, each has mean ability μ_j and effort productivity β_j . A fundamental is a J -vector $\omega := (\omega_j)_{j=1}^J$, where $\omega_j := (\mu_j, \beta_j)$. The true fundamental is ω^* . At the outset, each group j observes ω_j^* , and this information structure is common knowledge.

Society S, comprised of E and M, holds a prior over $\boldsymbol{\omega}$. Along dimension $\boldsymbol{\mu} := (\mu_j)_{j=1}^J$, the prior is degenerate at $\hat{\boldsymbol{\mu}} := (\hat{\mu}_j)_{j=1}^J$. Along dimension $\boldsymbol{\beta} := (\beta_j)_{j=1}^J$, the prior admits a density $p_0(\boldsymbol{\beta}) = \prod_{j=1}^J p_{0,j}(\beta_j)$, where each $p_{0,j}$ is smoothly supported on $[\underline{\beta}, \bar{\beta}]$.

In every period $n \in \mathbb{N}$, the following events unfold:

1. E commits to assess all groups at the same intensity $h_n \in [0, 1]$.
2. Groups privately and simultaneously make effort choices $a_{j,n} \geq 0$, $j = 1, \dots, J$.
3. Group j 's outcome is

$$X_{j,n} = \eta_{j,n} + r(a_{j,n}, \beta_j^*) + \varepsilon_{j,n},$$

where $\eta_{j,n} \sim \mathcal{N}(\mu_j^*, 1)$ and $\varepsilon_{j,n} \sim \mathcal{N}(0, h_n^{-1} - 1)$ denote the period- n ability and noise of the group. $\eta_{j,n} - \mu_j^*$ and $\varepsilon_{j,n}$ are distributed independently of one another, across groups, over time, and from $\boldsymbol{\omega}^*$. The vector of outcomes across groups, $\mathbf{X}_n := (X_{j,n})_{j=1}^n$, is publicly realized.

4. M rewards each group j for the expected value of $\eta_{j,n} + v_M(a_{j,n}, \beta_j)$.
5. S updates beliefs about $\boldsymbol{\beta}$.

All players are myopic and maximize expected payoffs period by period.

In period n , group j 's true effort choice is $a(h_n, \beta_j^*)$, although S perceives them as choosing $a(h_n, \beta_j)$. Let $H_n := (h_i, \mathbf{X}_i)_{i=1}^n$ denote the public history up to period n , and let $p_{j,n}(\cdot | H_n)$ denote the posterior density of β_j formed in Stage 5 of period n , conditional on H_n . In Stage 1 of period $n + 1$, E's assessment choice h_{n+1} solves

$$\max_{h \in [0,1]} \sum_{j=1}^J \alpha_j \int_{\underline{\beta}}^{\bar{\beta}} V_E(h, \beta_j) p_{j,n}(\beta_j | H_n) d\beta_j - \kappa(h). \quad (24)$$

The objective represents the expected payoffs from assessing all groups at the same intensity, weighted by their population, net of assessment cost.

Turning to belief updating: in Stage 5 of period n , the posterior density of β_j conditional on H_n is

$$p_{j,n}(\beta_j | H_n) = \frac{\exp \left(- \sum_{i=1}^n \frac{(X_{j,i} - \hat{\mu}_j - R_j(h_i, \beta_j))^2}{2h_i^{-1}} \right) p_{j,0}(\beta_j)}{\int_{\underline{\beta}}^{\bar{\beta}} \exp \left(- \sum_{i=1}^n \frac{(X_{j,i} - \hat{\mu}_j - R_j(h_i, \tilde{\beta}_j))^2}{2h_i^{-1}} \right) p_{j,0}(\tilde{\beta}_j) d\tilde{\beta}_j}.$$

The joint density $p_n(\boldsymbol{\beta} | H_n)$ equals the product of the marginal densities across groups.

Theorem 6. *Let p_0 be as described above, and suppose that Assumptions 1, 5, and 6 holds. As $n \rightarrow \infty$, S 's posterior beliefs about β converge to the Berk-NE vector $\hat{\beta}$ in Theorem 3 with positive probability, and with probability arbitrarily close to one given a sufficiently concentrated prior around $\hat{\beta}$.*

In contrast to Theorem 4, here we cannot establish a.s. convergence to the Berk-NE belief vector because the belief space is multidimensional. Beyond the one-dimensional case studied in Appendix C, the limit set of the ODE may include additional objects beyond steady states, such as cycles or even strange attractors. These objects are challenging to characterize without strong disciplines on the underlying geometry. Given this challenge, we focus on establishing positive-probability convergence to the Berk-NE vector $\hat{\beta}$, which is the unique steady state of the ODE and is also locally asymptotically stable.

The proof presented below assumes a uniform prior; the extension to general priors is analogous to that pursued in Appendix C.3 and is therefore omitted.

Under the uniform prior, posterior beliefs about β_j at the end of period n follow a truncated normal distribution $\tilde{\mathcal{N}}(m_{j,n}, v_{j,n})$, where $m_{j,n}$ and $v_{j,n}$ are determined as in (13) and (14). Posteriors are independent across groups. Write $\xi_{j,n} := 1/(nv_{j,n})$ and $\theta_{j,n} := (m_{j,n}, \xi_{j,n})$, and call $\theta_n := (\theta_{j,n})_{j=1}^J$ the period- n state. In period $n+1$, E 's optimal assessment solving (24) is fully parameterized by n and θ_n . We denote it by $h_{n+1}(\theta_n)$ and abbreviate it as h_{n+1} . For group j , the Fisher information of period- n outcome $X_{j,n}$ as a signal of β_j is $g_{2,j}^2(h_n)h_n := I_j(h_n)$.

A line-by-line replication of Appendix C.2 yields the update equation as

$$\theta_{n+1} - \theta_n = \frac{1}{n+1} \left(F_{n+1}(\theta_n) + b_{n+1}(\theta_n) \odot \epsilon_{n+1} \right), \quad (25)$$

where

$$F_{n+1}(\theta) := \begin{bmatrix} \frac{(I_1 \circ h_{n+1})(\theta) [(\beta_1 \circ h_{n+1})(\theta) - m_1]}{\frac{n}{n+1}\xi_1 + \frac{1}{n+1}(I_1 \circ h_{n+1})(\theta)} \\ \vdots \\ \frac{(I_J \circ h_{n+1})(\theta) [(\beta_J \circ h_{n+1})(\theta) - m_J]}{\frac{n}{n+1}\xi_J + \frac{1}{n+1}(I_J \circ h_{n+1})(\theta)} \\ (I_1 \circ h_{n+1})(\theta) - \xi_1 \\ \vdots \\ (I_J \circ h_{n+1})(\theta) - \xi_J \end{bmatrix},$$

$$b_{n+1}(\boldsymbol{\theta}) := \begin{bmatrix} \frac{\sqrt{(I_1 \circ h_{n+1})(\boldsymbol{\theta})}}{\frac{n}{n+1}\xi_1 + \frac{1}{n+1}(I_1 \circ h_{n+1})(\boldsymbol{\theta})} \\ \vdots \\ \frac{\sqrt{(I_J \circ h_{n+1})(\boldsymbol{\theta})}}{\frac{n}{n+1}\xi_J + \frac{1}{n+1}(I_J \circ h_{n+1})(\boldsymbol{\theta})} \\ \mathbf{0}_J \end{bmatrix}, \quad \boldsymbol{\epsilon}_{n+1} := \begin{bmatrix} \epsilon_{1,n+1} \\ \vdots \\ \epsilon_{J,n+1} \\ \mathbf{0}_J \end{bmatrix},$$

and $\odot$ denotes the element-wise product.

For large n , $\boldsymbol{\theta}_n$ becomes almost degenerate at $\tilde{\mathbf{m}}_n := (\tilde{m}_{j,n})_{j=1}^J$. The optimal assessment given $\delta_{\tilde{\mathbf{m}}_n}$ is $h(\tilde{\mathbf{m}}_n)$, where $h(\cdot)$ is the optimal assessment strategy in the equilibrium model that solves (5). In (25), drop $\boldsymbol{\epsilon}_n$, replace $h_{n+1}(\boldsymbol{\theta}_n)$ with $h(\tilde{\mathbf{m}}_n)$, and let $n \rightarrow \infty$. This gives rise to an ODE system:

$$\dot{\boldsymbol{\theta}} = F(\boldsymbol{\theta}), \quad (26)$$

where

$$F(\boldsymbol{\theta}) := \begin{bmatrix} \frac{(I_1 \circ h)(\tilde{\mathbf{m}}) [(\beta_1 \circ h)(\tilde{\mathbf{m}}) - m_1]}{\xi_1} \\ \vdots \\ \frac{(I_J \circ h)(\tilde{\mathbf{m}}) [(\beta_J \circ h)(\tilde{\mathbf{m}}) - m_J]}{\xi_J} \\ (I_1 \circ h)(\tilde{\mathbf{m}}) - \xi_1 \\ \vdots \\ (I_J \circ h)(\tilde{\mathbf{m}}) - \xi_J \end{bmatrix}.$$

For small ability misspecifications, F admits a unique steady state $\hat{\boldsymbol{\theta}} = (\hat{\boldsymbol{\beta}}, \hat{\boldsymbol{\xi}})$: $\hat{\boldsymbol{\xi}}$ is fully determined by $\hat{\boldsymbol{\beta}}$, and $\hat{\boldsymbol{\beta}}$ was solved in the proof of Theorem 3 and shown to be close to $\boldsymbol{\beta}^*$; therefore $\tilde{\boldsymbol{\beta}} = \hat{\boldsymbol{\beta}}$. Linearizing F around $\hat{\boldsymbol{\theta}}$ yields

$$\nabla F|_{\boldsymbol{\theta}=\hat{\boldsymbol{\theta}}} = -Id_{2J} + \left[\frac{\nabla_{\mathbf{m}}\boldsymbol{\psi}}{(\frac{d}{dh}\mathbf{I})(\nabla_{\mathbf{m}}h)^\top} \middle| \begin{array}{c} \mathbf{0}_{J \times J} \\ \mathbf{0}_{J \times J} \end{array} \right] \bigg|_{\boldsymbol{\theta}=\hat{\boldsymbol{\theta}}},$$

whose spectrum consists of the eigenvalue -1 (with multiplicity J) and the the spectrum of $-Id_J + \nabla_{\mathbf{m}}\boldsymbol{\psi}|_{\mathbf{m}=\hat{\boldsymbol{\beta}}} := \nabla_{\mathbf{m}}\boldsymbol{\Psi}|_{\mathbf{m}=\hat{\boldsymbol{\beta}}}$. The latter was shown in Appendix B to comprise of -1 (with multiplicity $J-1$) and $-1 + (\nabla_{\boldsymbol{\beta}}h)^\top \mathbf{g}(\boldsymbol{\beta}; \boldsymbol{\Delta})$. Since all eigenvalues are real and negative, $\hat{\boldsymbol{\theta}}$ is a sink and therefore a point attractor. Theorem 6 then follows by appealing to B99's attractor theorem; details are the exact same as in Appendix C.